Practical Multilayer Network Analysis with Py3plex

Release 1.0.1

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CONTENTS

CHAPTER

ONE

FRONT MATTER

1.1 About This Book

Practical Multilayer Network Analysis with Py3plex is a research-oriented handbook for graduate students and applied network scientists. It provides both theoretical foundations and practical guidance for analyzing complex systems using multilayer network representations.

This book covers:

- Mathematical foundations of multilayer networks
- The py3plex library architecture and design philosophy
- Hands-on examples and workflows for real network analysis
- **A powerful SQL-like DSL for network queries**
- Production deployment and reproducibility practices

DSL Feature Highlight

Py3plex includes a first-class DSL for querying networks with SQL-like syntax:

```
from py3plex.dsl import Q, L

# Find top influencers using builder API
result = (
    Q.nodes()
     .from_layers(L["social"])
     .where(degree__gt=5)
     .compute("betweenness_centrality")
     .order_by("-betweenness_centrality")
     .limit(10)
     .execute(network)
)
```

See **Part III** for complete DSL coverage!

1.2 Target Audience

This book is designed for:

- **Graduate students** in network science, computer science, and related fields
- **Applied researchers** working with complex relational data

- **Data scientists** analyzing social, biological, or infrastructure networks
- **Software engineers** building network analysis pipelines

1.2.1 Prerequisites

Readers should be familiar with:

- Python programming (intermediate level)
- Basic graph theory concepts
- Fundamental linear algebra

1.3 About the Software

py3plex (version 1.x) is an open-source Python library for multilayer and multiplex network analysis. It provides:

- Native support for multilayer network structures
- 17+ specialized algorithms for multilayer analysis
- A SQL-like query language (DSL) for network exploration
- NetworkX compatibility and interoperability
- High-performance I/O with Arrow/Parquet support

The library is actively maintained and tested on Python 3.8+.

1.4 How to Use This Book

Part I establishes foundations: what multilayer networks are, when to use them, and how py3plex is designed.

Part II teaches practical usage: installation, data loading, visualization, and running core algorithms.

Part III presents the DSL query language, a major feature for expressing complex analyses concisely.

Part IV demonstrates real-world applications through detailed case studies.

Part V covers systems topics: testing, reproducibility, and the web GUI.

Appendices provide reference material, detailed configurations, and extensive validation examples that supplement the main text.

1.4.1 Code Examples

All code examples in this book are runnable and tested. Examples use consistent naming and follow modern Python conventions. File paths reference the accompanying GitHub repository:

https://github.com/SkBlaz/py3plex

1.4.2 Conventions

- **Code blocks** use syntax highlighting and are self-contained
- **Mathematical notation** follows standard network science conventions
- **Feature status** is clearly marked:
  - **Stable** - Production-ready with guarantees
  - **Experimental** - Functional but may change

- **Planned** - Future features (mentioned only briefly)

1.5 Acknowledgments

This work builds on contributions from the py3plex community and research collaborations in multilayer network analysis.

If you use py3plex in your research, please cite:

```
@Article{Skrlj2019,
  author={Skrlj, Blaz and Kralj, Jan and Lavrac, Nada},
  title={Py3plex toolkit for visualization and analysis of multilayer networks},
  journal={Applied Network Science},
  year={2019},
  volume={4},
  number={1},
  pages={94},
  doi={10.1007/s41109-019-0203-7}
}
```

1.6 License

1.7 Version Information

- **Book version:** 1.0 (2025)
- **py3plex version:** 1.x
- **Python support:** 3.8+

1.7.1 Introduction & Motivation

Why Multilayer Networks?

Real-world systems rarely consist of a single type of relationship. Consider:

- **Social networks:** People interact through friendship, collaboration, family ties, and online connections—each relationship type has different meanings and dynamics.
- **Transportation systems:** Cities connect via air, rail, and road networks. A disruption in one mode affects the others, and travelers make choices across modes.
- **Biological systems:** Proteins interact through physical binding, regulatory relationships, and metabolic pathways—each layer represents a different mechanism.
- **Information systems:** Software components depend on each other through API calls, data flows, and deployment relationships.

Traditional single-layer (monoplex) networks flatten these diverse relationships into a single graph, losing critical structural information. **Multilayer networks** preserve the distinct nature of different relationship types by organizing them into separate but interconnected layers.

The Cost of Flattening

When we flatten a multilayer network into a single graph, we lose:

1. **Relationship semantics:** A coauthor edge is fundamentally different from a Twitter follower edge, but flattening treats them identically.
2. **Community structure:** Groups that are well-separated in individual layers may appear spuriously connected when layers are combined.
3. **Node roles:** A person might be central in their professional network but peripheral in their hobby network—this context-dependent centrality disappears upon flattening.
4. **Cross-layer dynamics:** Information spreading from email to in-person meetings follows specific patterns that become invisible when layers are merged.

Example: A researcher with 2 coauthors and 500 Twitter followers has degree 502 in a flattened network, suggesting high influence. But their academic impact (degree 2) is actually modest—the Twitter followers don't write papers with them. Multilayer analysis preserves this distinction.

What py3plex Enables

py3plex is a Python library designed for practical analysis of multilayer networks. It provides:

1. Expressive Data Structures

Native support for multilayer network representations that mirror how real systems are naturally structured:

```
from py3plex.core import multinet

# Create a multilayer network
network = multinet.multi_layer_network()

# Add edges across multiple layers
network.add_edges([
    ['Alice', 'coauthor', 'Bob', 'coauthor', 1],
    ['Alice', 'twitter', 'Carol', 'twitter', 1],
    ['Bob', 'coauthor', 'Dave', 'coauthor', 1],
], input_type="list")
```

2. Multilayer-Aware Algorithms

Algorithms specifically designed for multilayer structure:

- **Community detection** that respects layer boundaries while finding cross-layer modules
- **Centrality measures** that account for both intra-layer and inter-layer importance
- **Random walks** that can transition between layers
- **Dynamics models** (e.g., epidemic spreading) with layer-specific parameters

3. A Query Language for Networks

A SQL-like DSL (Domain-Specific Language) for expressing complex network analyses concisely:

DSL Example: Concise Network Analysis

Instead of writing loops and conditions, use declarative queries:

```python
from py3plex.dsl import Q, L

# Find high-degree nodes in the coauthor layer
result = (
    Q.nodes()
     .from_layers(L["coauthor"])
     .where(degree__gt=5)
     .compute("betweenness_centrality", "pagerank")
     .order_by("-betweenness_centrality")
     .limit(10)
     .execute(network)
)

# Export for further analysis
result.to_pandas().to_csv("top_researchers.csv")
```

This replaces dozens of lines of manual filtering and iteration with a clear, composable query.

Key DSL features:

- SQL-like syntax familiar to data analysts
- Layer algebra: `L["social"] + L["work"] - L["bots"]`
- Type-safe Python builder API
- Export to pandas, NetworkX, Arrow, CSV, JSON
- EXPLAIN mode for query optimization

4. Visualization for Multilayer Networks

Publication-ready visualizations that show layer structure and cross-layer connections:

```python
# Visualize with layer separation
network.visualize_network(
    show=True,
    layout_style="force_directed_3d"
)
```

5. Production-Ready Infrastructure

- **High-performance I/O** with Arrow/Parquet support for large networks
- **NetworkX compatibility** for interoperability with the broader Python ecosystem
- **Testing and validation** frameworks to ensure correctness
- **Docker containers** and CLI tools for deployment

Who Should Use This Book

This book is designed for:

Graduate Students and Researchers

If you're analyzing social, biological, or infrastructure networks in your research, this book provides both theoretical foundations and practical tools. You'll learn when multilayer modeling is appropriate and how to apply specialized algorithms correctly.

Data Scientists and Analysts

If you're working with complex relational data—user behavior across platforms, supply chain networks, knowledge graphs—py3plex offers a principled way to preserve and analyze structural nuances.

Software Engineers

If you're building network analysis pipelines, this book shows how to use py3plex's APIs, DSL, and deployment tools to create maintainable, efficient systems.

Prerequisites

This book assumes:

- **Intermediate Python** (classes, list comprehensions, basic NumPy)
- **Basic graph theory** (nodes, edges, paths, degree)
- **Fundamental linear algebra** (matrices, matrix multiplication)

You don't need prior multilayer network experience—Part I builds from fundamentals.

Real-World Applications

Multilayer network analysis has proven valuable in numerous domains:

Biological Networks

- Protein-protein interactions across multiple tissues or conditions
- Gene regulatory networks with transcription, translation, and metabolic layers
- Brain connectivity with structural and functional layers

Social Networks

- User behavior across multiple social platforms
- Organizational networks with formal and informal relationships
- Citation networks with papers, authors, and venues

Infrastructure Networks

- Transportation with air, rail, and road connections
- Power grids with generation, transmission, and distribution layers
- Communication networks with physical and logical layers

Knowledge and Information

- Heterogeneous information networks (authors-papers-venues)
- Semantic networks with multiple relationship types
- Code dependency graphs with API, data, and deployment layers

Each of these applications benefits from multilayer analysis by capturing the true structure of the system rather than forcing it into a single-layer representation.

How This Book Is Organized

Part I (Chapters 1-3): Foundations
Introduces multilayer networks, defines key concepts, and explains py3plex's design philosophy.

Part II (Chapters 4-7): Working with Py3plex
Covers installation, data loading, visualization, and core algorithms—everything needed for basic usage.

Part III (Chapters 8-11): Advanced Analysis and the DSL
Deep dive into the query language, a major feature for expressing complex analyses concisely.

Part IV (Chapters 12-14): Case Studies
Complete, reproducible analyses of real datasets across different domains.

Part V (Chapters 15-17): Systems and Deployment
Testing, reproducibility, and production deployment including the web GUI.

Appendices (A-E): Reference Material
Detailed configurations, validation scripts, error handling, and API reference.

Each chapter includes:

- **Conceptual explanations** with mathematical foundations where needed
- **Code examples** that are runnable and tested
- **Practical tips** based on real usage patterns
- **References** for deeper study

Summary

Multilayer networks provide a natural and expressive way to model systems with multiple types of relationships. By preserving the distinct nature of different layers, we can:

- Analyze structure more accurately
- Compute meaningful centralities and communities
- Model dynamics more realistically
- Make better predictions and interventions

py3plex makes multilayer network analysis accessible and practical, with a focus on:

- Expressive data structures
- Specialized algorithms
- A powerful query language
- Production-ready tools

The following chapters will show you how to use these capabilities effectively.

Further Reading

- Kivelä, M., et al. (2014). "Multilayer networks." *Journal of Complex Networks* 2(3): 203-271. [Comprehensive review of multilayer network theory]
- Boccaletti, S., et al. (2014). "The structure and dynamics of multilayer networks." *Physics Reports* 544(1): 1-122. [Physics-focused perspective]
- Bianconi, G. (2018). *Multilayer Networks: Structure and Function.* Oxford University Press. [Textbook treatment]

1.7.2 Multilayer Network Basics

This chapter establishes the formal foundations of multilayer networks. We define key concepts, introduce mathematical notation, and show how different types of multilayer networks relate to each other.

Formal Definitions

Node-Layer Pairs

A **multilayer network** $\mathcal{M}$ consists of:

- A set of **nodes** $V = \{v_1, v_2, ..., v_N\}$
- A set of **layers** $L = \{\alpha, \beta, \gamma, ...\}$
- A set of **node-layer pairs** $V_M = V \times L$
- **Intra-layer edges** $E_\alpha \subseteq V \times V$ within each layer α
- **Inter-layer edges** $E_C \subseteq V_M \times V_M$ connecting node-layer pairs

The **node-layer pair** (v, α) is the fundamental unit: it represents node v in the context of layer α.

Supra-Adjacency Matrix

The **supra-adjacency matrix** $\mathbf{A}$ is a block matrix that encodes the full multilayer structure:

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{\alpha\alpha} & \mathbf{A}_{\alpha\beta} & \cdots \\ \mathbf{A}_{\beta\alpha} & \mathbf{A}_{\beta\beta} & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

Where:

- **Diagonal blocks** $\mathbf{A}_{\alpha\alpha}$ are the adjacency matrices of individual layers
- **Off-diagonal blocks** $\mathbf{A}_{\alpha\beta}$ encode inter-layer connections

For a multiplex network with N nodes and L layers, $\mathbf{A}$ is an $(N \times L) \times (N \times L)$ matrix.

Example: A 3-node, 2-layer network:

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{\text{friends}} & \omega\mathbf{I} \\ \omega\mathbf{I} & \mathbf{A}_{\text{colleagues}} \end{pmatrix}$$

where ω is the **inter-layer coupling strength** and $\mathbf{I}$ is the identity matrix (representing identity edges: each node connects to itself across layers).

Types of Multilayer Networks

Multiplex Networks

Definition: A multiplex network has the same node set in all layers, with different edge sets per layer.

$$V_\alpha = V_\beta = \cdots = V \quad \text{for all layers}$$

Characteristics:

- **Same entities** appear in all layers
- **Different relationship types** per layer
- **Strong inter-layer coupling** (typically $\omega = 1.0$ for identity edges)

Example: Social Multiplex Network

```
from py3plex.core import multinet

# Create multiplex network
network = multinet.multi_layer_network(network_type="multiplex")

# Same people, different relationship types
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Bob', 'friends', 'Carol', 'friends', 1],
    ['Alice', 'colleagues', 'Bob', 'colleagues', 1],
    ['Bob', 'colleagues', 'Dave', 'colleagues', 1],
], input_type="list")

# Verify structure
print(f"Layers: {network.get_layers()}")
print(f"Nodes per layer: {network.get_number_of_nodes_per_layer()}")
```

Real-world applications:

- Social networks across platforms (Facebook, Twitter, LinkedIn)
- Transportation modes (air, rail, road)
- Communication channels (email, phone, chat)
- Biological interactions (genetic, protein, metabolic)

Heterogeneous Information Networks

Definition: A network with different node types, where edges connect specific node type pairs.

Characteristics:

- **Different node types** per layer (authors, papers, venues)
- **Type-specific edges** (author-paper, paper-venue)
- **No inter-layer coupling** (nodes don't repeat across layers)

Example: Academic Network

```
network = multinet.multi_layer_network()

# Different node types
network.add_edges([
    ['Alice', 'authors', 'Paper1', 'papers', 1],
    ['Bob', 'authors', 'Paper1', 'papers', 1],
    ['Paper1', 'papers', 'ICML', 'venues', 1],
    ['Paper2', 'papers', 'ICML', 'venues', 1],
], input_type="list")
```

This creates a **tripartite** network with three node types: authors, papers, and venues.

Real-world applications:

- Academic networks (authors, papers, venues, keywords)
- E-commerce (users, products, sellers, categories)
- Biomedical (drugs, diseases, targets, pathways)

- Knowledge graphs (entities, relations, concepts)

Temporal Networks

Definition: Networks that evolve over time, represented as time-sliced layers.

Characteristics:

- **Time windows as layers** (2020, 2021, 2022, ...)
- **Node presence varies** across time slices
- **Temporal edges** connect adjacent time slices

Example: Temporal Social Network

```
network = multinet.multi_layer_network()

# Time-sliced layers
network.add_edges([
    ['Alice', '2020', 'Bob', '2020', 1],
    ['Alice', '2021', 'Bob', '2021', 1],
    ['Bob', '2021', 'Carol', '2021', 1],
    ['Alice', '2022', 'Carol', '2022', 1],
], input_type="list")
```

Real-world applications:

- Communication patterns over time
- Disease spread through populations
- Financial transaction networks
- Collaboration evolution

Interdependent Networks

Definition: Multiple networks where the function of nodes in one layer depends on nodes in other layers.

Characteristics:

- **Dependency edges** encode functional relationships
- **Cascading failures** possible across layers
- **Critical infrastructure** applications

Example: Power grid (layer 1) depends on communication network (layer 2). If communication fails, power grid control fails.

Real-world applications:

- Infrastructure systems (power, water, communication)
- Supply chain networks
- Cyber-physical systems
- Ecological networks (species, habitats, resources)

Inter-Layer Coupling

The **coupling strength** ω controls how strongly layers are connected.

Identity Coupling

The most common form connects each node to itself across layers:

$$E_C = \{((v, \alpha), (v, \beta)) : v \in V, \alpha \neq \beta \in L\}$$

with edge weight ω.

Choice of ω:

- $\omega = 1.0$ — Layers are equally important, nodes fully correspond
- $\omega < 1.0$ — Layers are loosely coupled, favor intra-layer paths
- $\omega > 1.0$ — Inter-layer transitions are encouraged

Example:

```
network = multinet.multi_layer_network()

# Add intra-layer edges
network.add_edges([
    ['A', 'layer1', 'B', 'layer1', 1],
    ['A', 'layer2', 'C', 'layer2', 1],
], input_type="list")

# Add inter-layer coupling (identity edges)
network.add_edges([
    ['A', 'layer1', 'A', 'layer2', 0.5],  # omega = 0.5
], input_type="list")
```

General Coupling

More complex couplings allow different nodes to connect across layers:

$$E_C = \{((v, \alpha), (w, \beta)) : (v, w) \in C_{\alpha\beta}\}$$

where $C_{\alpha\beta}$ defines which nodes couple between layers α and β.

When to Use Multilayer Networks

Use multilayer modeling when:

1. **Multiple relationship types have distinct semantics**

   Example: Friendship vs. professional collaboration—these networks have different properties and dynamics.

2. **Node roles vary by context**

   Example: A person central in academic network may be peripheral in social media network.

3. **Cross-layer interactions matter**

   Example: Information spreads from Twitter to traditional media—this cross-layer flow is meaningful.

4. **Temporal evolution is important**

   Example: Community structure evolves over time—time-sliced layers preserve this evolution.

5. **System-level resilience depends on layer dependencies**

   Example: Power grid failure affects communication, which affects emergency response.

Choosing a Modeling Approach

Decision Tree

```
1. Do relationships have fundamentally different types?
   YES → Use layers (multiplex or heterogeneous)
   NO  → Use edge weights or attributes

2. Are the same entities present in all layers?
   YES → Multiplex network (strong coupling)
   NO  → Heterogeneous network (weak/no coupling)

3. Does time evolution matter?
   YES → Temporal layers (time-sliced)
   NO  → Static multilayer network

4. Are there functional dependencies between layers?
   YES → Interdependent network (dependency edges)
   NO  → Standard multiplex/heterogeneous
```

Common Mistakes

Over-aggregation
: Combining layers that should stay separate (e.g., merging email and meeting networks loses information about mode transitions).

Under-aggregation
: Creating too many sparse layers when edge weights would suffice (e.g., separate layer for each email vs. email timestamp as attribute).

Wrong coupling strength
: Using $\omega = 1.0$ when nodes don't fully correspond, or $\omega \to 0$ when they do.

Mismatched identifiers
: Using different IDs for the same entity across layers breaks multiplex structure.

Why Flattening Fails

Flattening (aggregating all layers into a single graph) loses critical information:

1. Community Structure

Multilayer communities may have:

- **Core in one layer, periphery in another**
- **Cross-layer bridges** that appear as spurious intra-layer connections when flattened

Example: Academic collaboration (dense group) + Twitter followers (different dense group). Flattening creates false bridges.

2. Centrality

A node's importance depends on layer:

- **Structural centrality** in one layer (high degree)

- **Functional centrality** in another (betweenness for information flow)

Flattening mixes these distinct roles.

3. Path Structure

Meaningful paths often cross layers:

- Email → meeting → collaboration
- Twitter mention → news coverage → policy change

Flattening hides these cross-layer transitions.

4. Dynamics

Disease spreading, information diffusion, and cascading failures all depend on layer-specific parameters and inter-layer transitions. Flattening cannot capture:

- **Layer-specific transmission rates**
- **Mode-switching dynamics**
- **Asymmetric cross-layer effects**

Key Terminology

Intra-layer edges
Edges within a single layer, connecting (v, α) to (w, α).

Inter-layer edges
Edges between layers, connecting (v, α) to (w, β) with $\alpha \neq \beta$.

Node-layer pair
The tuple (v, α) representing node v in layer α—the fundamental unit of a multilayer network.

Supra-adjacency matrix
The block matrix $\mathbf{A}$ encoding all intra-layer and inter-layer edges.

Coupling strength
The weight ω of inter-layer edges, controlling how strongly layers interact.

Multiplex network
Multilayer network with the same nodes in all layers.

Heterogeneous information network
Multilayer network with different node types per layer.

Working with Supra-Adjacency Matrices in Py3plex

```
from py3plex.core import multinet, random_generators

# Generate random multilayer network
network = random_generators.random_multilayer_ER(
    num_nodes=100,
    num_layers=3,
    probability=0.05,
    directed=False
)

# Get supra-adjacency matrix (sparse format)
supra_matrix = network.get_supra_adjacency_matrix()
```

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```
print(f"Matrix shape: {supra_matrix.shape}")  # (300, 300) for 100 nodes × 3 layers
print(f"Matrix density: {supra_matrix.nnz / (supra_matrix.shape[0] ** 2):.4f}")

# Visualize matrix structure
network.visualize_matrix({"display": True})
```

The supra-adjacency matrix enables tensor-based algorithms and linear algebra operations on the full multilayer structure.

Summary

Multilayer networks formalize systems with multiple relationship types by:

- Using **node-layer pairs** as the fundamental unit
- Encoding structure in the **supra-adjacency matrix**
- Supporting various types: **multiplex**, **heterogeneous**, **temporal**, **interdependent**
- Controlling interaction via **coupling strength** ω

Key takeaways:

1. **Multiplex** = same nodes, different edge types
2. **Heterogeneous** = different node types per layer
3. **Temporal** = time-sliced layers
4. **Coupling** = inter-layer connections (identity edges most common)
5. **Flattening loses information** — use multilayer analysis when layer structure matters

The next chapter explains how py3plex implements these concepts and why its design choices support efficient, correct multilayer analysis.

Further Reading

- **Theory:** Kivelä et al. (2014). "Multilayer networks." *J. Complex Networks* 2(3): 203-271.
- **Physics:** Boccaletti et al. (2014). "The structure and dynamics of multilayer networks." *Physics Reports* 544(1): 1-122.
- **Textbook:** Bianconi, G. (2018). *Multilayer Networks: Structure and Function.* Oxford University Press.
- **Applications:** De Domenico et al. (2013). "Mathematical formulation of multilayer networks." *Physical Review X* 3(4): 041022.

1.7.3 Design of Py3plex

This chapter explains how py3plex is architected and why it makes specific design choices. Understanding these design principles will help you use the library more effectively and debug unexpected behavior.

Architecture Overview

Py3plex is built on three foundational principles:

1. **NetworkX compatibility** — Leverage a mature, well-tested ecosystem
2. **Modular design** — Separate concerns for flexibility and extensibility

3. **Simple, off-the-shelf functionality** — Minimize barrier to entry

The library consists of several core modules:

```
py3plex/
├── core/                    # Data structures (multi_layer_network class)
├── algorithms/              # Analysis methods
│   ├── statistics/          # Multilayer statistics
│   ├── centrality/          # Centrality measures
│   ├── community_detection/ # Community algorithms
│   └── paths/               # Path algorithms
├── dynamics/                # Dynamical processes (SIS, SIR, random walks)
├── dsl/                     # SQL-like query language
├── visualization/           # Plotting and visualization
├── io/                      # Data loading/saving
└── cli.py                   # Command-line interface
```

Node-Layer Pair Representation

The Fundamental Unit

Py3plex represents multilayer networks using **node-layer pairs** as the fundamental unit. A node v in layer α is stored as the tuple (v, α).

```python
from py3plex.core import multinet

network = multinet.multi_layer_network()

# Add nodes explicitly as (node_id, layer_id) pairs
network.add_nodes([
    ('Alice', 'friends'),
    ('Bob', 'friends'),
    ('Alice', 'colleagues'),
])

# Or implicitly through edge addition
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Alice', 'colleagues', 'Bob', 'colleagues', 1],
], input_type="list")
```

Why node-layer pairs?

- **Unambiguous identification** — The same logical entity (Alice) has distinct identities in different layers
- **NetworkX compatibility** — Tuples are valid NetworkX node identifiers
- **Efficient lookups** — Python's tuple hashing enables fast operations
- **Natural tensor mapping** — Node-layer pairs map directly to rows/columns of the supra-adjacency matrix

The `multi_layer_network` Class

Core Components

The `multi_layer_network` class wraps a NetworkX graph and adds multilayer-specific functionality:

```
network = multinet.multi_layer_network(directed=False)

# The underlying NetworkX graph (MultiGraph or MultiDiGraph)
G = network.core_network

# Layer management
layers = network.get_layers()           # ['friends', 'colleagues']
layer_map = network.layer_name_map      # {'friends': 0, 'colleagues': 1}

# Access node-layer pairs
nodes = list(network.get_nodes())       # [('Alice', 'friends'), ...]
```

Key attributes:

- `core_network` — The NetworkX graph storing all node-layer pairs and edges
- `layer_name_map` — Mapping from layer names to internal IDs
- `directed` — Whether the network is directed

Multilayer vs. Multiplex Mode

Py3plex supports two operational modes:

Multilayer (default):

- General multilayer networks with arbitrary node sets per layer
- No automatic inter-layer edges
- Use for heterogeneous networks (different node types)

```
# Heterogeneous network: authors and papers
network = multinet.multi_layer_network(network_type='multilayer')
network.add_edges([
    ['Alice', 'authors', 'Paper1', 'papers', 1],
    ['Paper1', 'papers', 'ICML', 'venues', 1],
], input_type="list")
```

Multiplex:

- Same node set in all layers
- Automatically creates identity (coupling) edges with `type='coupling'`
- Use when the same entities appear in all layers

```
# Social network: same people, different relationships
network = multinet.multi_layer_network(network_type='multiplex')
network.load_network('social.edges', input_type='multiplex_edges')

# Coupling edges are auto-created: (Alice, friends) <-> (Alice, colleagues)

# Get explicit edges only (exclude coupling)
explicit_edges = list(network.get_edges(multiplex_edges=False))
```

Relationship to NetworkX

Direct Access to NetworkX

Py3plex does not hide NetworkX—it exposes the underlying graph for direct manipulation:

```
import networkx as nx

# Access the graph directly
G = network.core_network

# Use any NetworkX function
betweenness = nx.betweenness_centrality(G)
communities = nx.community.louvain_communities(G)

# Modify the graph directly
G.nodes[('Alice', 'friends')]['age'] = 30
```

Why this matters:

- You can use **any NetworkX algorithm** on the multilayer network
- You can **extend py3plex** by writing functions that operate on `core_network`
- You don't need to learn a completely new API—if you know NetworkX, you know most of py3plex

NetworkX Interoperability

Convert between py3plex and NetworkX:

```
# From NetworkX to py3plex
import networkx as nx
G = nx.karate_club_graph()
network = multinet.multi_layer_network()
network.load_network_from_networkx(G, layer_name='social')

# From py3plex to NetworkX (extract a single layer)
friends_layer = network.get_layer_subgraph('friends')
```

This enables workflows that mix py3plex's multilayer capabilities with NetworkX's extensive algorithm library.

Core Modules

Algorithms

Statistics (`py3plex/algorithms/statistics/`)
: Multilayer-specific metrics like degree distributions, layer overlap, and aggregation statistics.

Centrality (`py3plex/algorithms/centrality/`)
: Centrality measures adapted for multilayer networks, including explainable centrality that breaks down scores by layer.

Community Detection (`py3plex/algorithms/community_detection/`)
: Multilayer Louvain, Infomap, and modularity-based methods.

Paths (`py3plex/paths/`)
: Shortest paths, random walks, and flow algorithms that respect layer structure.

Dynamics

Dynamical Processes (`py3plex/dynamics/`)
: OOP-style classes for epidemic models (SIS, SIR, SEIR), random walks, and custom dynamics. Each model has:

- `set_seed()` for reproducibility
- `run()` for execution
- `get_measure()` for extracting results

DSL and Query Language

Domain-Specific Language (`py3plex/dsl/`)
: SQL-like queries for network analysis:

```
from py3plex.dsl import Q, L

result = (
    Q.nodes()
     .from_layers(L["friends"])
     .where(degree__gt=5)
     .compute("betweenness_centrality")
     .execute(network)
)
```

The DSL provides:

- **Builder API** — Chainable, type-hinted query construction
- **String syntax** — SQL-like queries as strings
- **EXPLAIN mode** — Query execution plans
- **Export** — Results to pandas, JSON, CSV

Visualization

Visualization (`py3plex/visualization/`)
: Publication-ready plots including:

- **Hairball plots** — 2D/3D force-directed layouts with layer separation
- **Matrix plots** — Supra-adjacency matrix visualization
- **Diagonal layouts** — Specialized for large multilayer networks

I/O

Input/Output (`py3plex/io/`)
: Support for multiple formats:

- **Edgelists** — Simple text format
- **GraphML** — XML-based format
- **Arrow/Parquet** — High-performance columnar format for large networks
- **JSON** — Flexible structured format

Design Principles in Practice

1. Flexible Input

Multiple ways to accomplish the same task:

```
# Method 1: List format
network.add_edges([['A', 'L1', 'B', 'L1', 1]], input_type="list")

# Method 2: Dict format (explicit)
network.add_edges([{
    'source': 'A', 'target': 'B',
    'source_type': 'L1', 'target_type': 'L1'
}])

# Method 3: Load from file
network.load_network("data.edgelist", input_type="edgelist")
```

2. Lazy Evaluation

Expensive operations are computed only when needed:

```
# Network creation and edge addition are fast
network = multinet.multi_layer_network()
network.add_edges([...])  # No supra-adjacency matrix computed

# Matrix is computed only when requested
supra_adj = network.get_supra_adjacency_matrix()  # Computed here
```

3. Graceful Degradation

The library handles edge cases and provides informative warnings:

- Missing layers → created automatically
- Empty networks → return sensible defaults
- Invalid parameters → clear error messages

4. Extensibility

Adding new algorithms is straightforward:

```
def my_custom_metric(network):
    """Custom multilayer metric."""
    G = network.core_network
    # Implement your algorithm using G
    return result

# Use immediately
score = my_custom_metric(network)
```

Why These Choices Matter

NetworkX compatibility means you can:

- Use hundreds of existing algorithms without modification
- Mix py3plex with other NetworkX-based tools
- Leverage a mature, well-documented ecosystem

Node-layer pair representation ensures:

- Unambiguous node identity across layers
- Efficient graph operations (hashing, lookups)
- Direct mapping to mathematical definitions

Modular architecture allows:

- Importing only needed functionality
- Easy testing and debugging
- Clean separation of concerns

Multiple input methods accommodate:

- Different data sources and formats
- User preferences and workflows
- Programmatic and interactive use

Limitations and Trade-offs

No design is perfect. Py3plex makes conscious trade-offs:

Memory:
: The supra-adjacency matrix for large networks can be memory-intensive. Use sparse matrix formats (default) and avoid materializing the full matrix when possible.

Performance:
: Some operations (e.g., cross-layer random walks) require iteration over node-layer pairs, which is slower than pure NetworkX on single-layer graphs.

Coupling semantics:
: The automatic coupling in multiplex mode assumes identity edges. For more complex inter-layer relationships, use multilayer mode and add edges explicitly.

Summary

Py3plex is designed around:

1. **Node-layer pairs** as the fundamental unit
2. **NetworkX compatibility** for interoperability and extensibility
3. **Modular architecture** for flexibility
4. **Simple, practical APIs** for ease of use

These principles enable:

- Correct multilayer analysis (proper layer semantics)
- Efficient implementation (leverage NetworkX)

- Extensible design (add your own algorithms)
- Practical workflows (multiple input formats, visualization, DSL)

The following chapters will show how to use these design elements in practice: loading data, visualizing networks, and running multilayer-specific algorithms.

Further Reading

- NetworkX documentation: https://networkx.org/documentation/stable/
- Py3plex API reference: *Appendix E: Extended API and DSL Reference*
- Code architecture: *Appendix A: Repository Layout and Scripts*

1.7.4 Installation and Getting Started

This chapter covers installing py3plex and running your first multilayer network analysis. We provide a minimal quick-start example followed by installation details for different use cases.

DSL Quick Start

After installation, try the DSL for instant network exploration:

```
from py3plex.core import multinet
from py3plex.dsl import Q, L

# Create a simple network
network = multinet.multi_layer_network()
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Bob', 'friends', 'Carol', 'friends', 1],
], input_type="list")

# Query it with DSL
result = (
    Q.nodes()
     .compute("degree")
     .order_by("-degree")
     .execute(network)
)
print(result.to_pandas())
```

DSL makes network analysis intuitive from day one!

Quick Install

For most users, installation is straightforward:

```
pip install py3plex
```

This installs py3plex with core dependencies (NetworkX, NumPy, SciPy, pandas, matplotlib).

Verify installation:

```
import py3plex
print(py3plex.__version__)  # Should print 1.x
```

System Requirements

Python: 3.8 or higher (tested on 3.8, 3.9, 3.10, 3.11, 3.12)

Platforms: Linux, macOS, Windows

Core dependencies (installed automatically):

- NetworkX (2.8) — Graph data structures and algorithms
- NumPy (1.22) — Numerical computing and array operations
- SciPy (1.8) — Sparse matrices and scientific computing
- pandas (1.4) — Data manipulation and analysis
- matplotlib (3.5) — Visualization and plotting
- scikit-learn (1.0) — Machine learning utilities

Hello Multilayer Network

Create and analyze a multilayer network in 30 seconds:

```
from py3plex.core import multinet

# Create network
network = multinet.multi_layer_network()

# Add edges: [source, source_layer, target, target_layer, weight]
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Bob', 'friends', 'Carol', 'friends', 1],
    ['Alice', 'colleagues', 'Bob', 'colleagues', 1],
    ['Bob', 'colleagues', 'Dave', 'colleagues', 1],
], input_type="list")

# Display statistics
network.basic_stats()
```

Output:

```
Number of nodes: 6
Number of edges: 4
Number of unique nodes (as node-layer tuples): 6
Number of unique node IDs (across all layers): 4
Nodes per layer:
  Layer 'friends': 3 nodes
  Layer 'colleagues': 3 nodes
```

Key insight: The network has 6 node-layer pairs (Alice-friends, Alice-colleagues, Bob-friends, Bob-colleagues, Carol-friends, Dave-colleagues) but only 4 unique people.

Visualize the Network

```
from py3plex.visualization.multilayer import draw_multilayer_default
```

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```
# Simple visualization
draw_multilayer_default([network], display=True)
```

This creates a force-directed layout with nodes colored by layer.

Basic Analysis

```
from py3plex.algorithms.statistics import multilayer_statistics as mls

# Layer density (how connected is each layer?)
print(f"Friends density: {mls.layer_density(network, 'friends'):.3f}")
# Output: Friends density: 0.667 (2 of 3 possible edges exist)

# Node activity (fraction of layers a node appears in)
print(f"Bob's activity: {mls.node_activity(network, 'Bob'):.3f}")
# Output: Bob's activity: 1.000 (appears in both layers)
```

Query with DSL

Use SQL-like queries to explore the network:

```
from py3plex.dsl import execute_query

# Find high-degree nodes
result = execute_query(network, 'SELECT nodes WHERE degree > 1')
print(f"Found {result['count']} nodes with degree > 1")

# Get all nodes in a specific layer
result = execute_query(network, 'SELECT nodes WHERE layer="friends"')
```

The DSL (covered in Part III) provides a powerful way to filter, compute, and analyze multilayer networks without writing explicit loops.

Installation Options

Development Installation

For contributors or to access the latest features:

```
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex
pip install -e .
```

The `-e` flag installs in editable mode—changes to source code are immediately available.

For development with testing/linting tools:

```
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex
make setup        # Create virtual environment
make dev-install  # Install with dev dependencies
```

This installs pytest, black, ruff, mypy, and sphinx for development workflows.

Docker Installation

For a containerized environment with all dependencies:

```
# Clone repository
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex

# Build image
docker build -t py3plex:latest .

# Run commands
docker run --rm py3plex:latest --version
```

Benefits:

- No Python setup required
- Consistent environment across systems
- Isolated from system Python

Optional Dependencies

Py3plex supports optional feature sets:

```
# Advanced community detection (Infomap)
pip install py3plex[infomap]

# Extra algorithms (Louvain, cdlib)
pip install py3plex[algos]

# Advanced visualization (Plotly, igraph)
pip install py3plex[viz]

# High-performance I/O (Arrow/Parquet)
pip install py3plex[arrow]

# Install multiple extras
pip install py3plex[infomap,viz,algos,arrow]
```

Virtual Environments

We strongly recommend using virtual environments:

Using venv:

```
python3 -m venv py3plex-env
source py3plex-env/bin/activate  # Linux/macOS
# py3plex-env\Scripts\activate   # Windows
pip install py3plex
```

Using conda:

```
conda create -n py3plex python=3.10
conda activate py3plex
pip install py3plex
```

Key Concepts

Before proceeding, understand these fundamental concepts:

Node-Layer Pairs

In py3plex, a node is always associated with a layer. The tuple `(node_id, layer_id)` is the fundamental unit.

- `('Alice', 'friends')` is different from `('Alice', 'colleagues')`
- This allows the same logical entity to have different properties and connections in different contexts

Example:

```
# Access nodes as (node_id, layer_id) tuples
for node in network.get_nodes():
    print(node)  # ('Alice', 'friends'), ('Bob', 'friends'), ...
```

NetworkX Compatibility

Py3plex uses NetworkX as its underlying graph representation:

```
import networkx as nx

# Direct access to the NetworkX graph
G = network.core_network

# Use any NetworkX algorithm
betweenness = nx.betweenness_centrality(G)

# Modify directly
G.nodes[('Alice', 'friends')]['age'] = 30
```

This means:

- Any NetworkX algorithm works on py3plex networks
- You can mix py3plex and NetworkX functions freely
- Learning curve is minimal if you already know NetworkX

Input Formats

Py3plex supports multiple edge input formats:

List format (used in examples above):

```
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
], input_type="list")
```

Dictionary format (explicit, Pythonic):

```
network.add_edges([{
    'source': 'Alice',
    'source_type': 'friends',
    'target': 'Bob',
    'target_type': 'friends',
    'weight': 1
}])
```

File loading (for larger networks):

```
# Edgelist format
network.load_network("data.edgelist", input_type="edgelist")

# GraphML, JSON, Arrow, etc.
# See Chapter 5 for details
```

Common Issues

Issue: "No module named 'py3plex'"

Solution: Ensure py3plex is installed in the active Python environment:

```
pip install py3plex
```

Verify which Python interpreter you're using:

```
which python  # Linux/macOS
where python  # Windows
```

Issue: Import errors for optional dependencies

Solution: Install the relevant extra:

```
pip install py3plex[infomap]  # For Infomap
pip install py3plex[viz]      # For Plotly
```

Issue: Permission denied on Linux/macOS

Solution: Use a virtual environment (recommended) or install with `--user`:

```
pip install --user py3plex
```

Verifying Installation

Run a self-test to confirm everything works:

```
from py3plex.core import multinet
from py3plex.algorithms.statistics import multilayer_statistics as mls

# Create test network
network = multinet.multi_layer_network()
network.add_edges([
    ['A', 'L1', 'B', 'L1', 1],
```

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```
    ['B', 'L1', 'C', 'L1', 1],
], input_type="list")

# Test basic operations
assert network.get_number_of_nodes() == 3
assert mls.layer_density(network, 'L1') > 0

print("✓ py3plex is working correctly")
```

Next Steps

Now that you have py3plex installed and understand the basics:

- **Chapter 5** — Learn about data loading, supported formats, and best practices for representing multilayer networks
- **Chapter 6** — Explore visualization techniques for multilayer networks
- **Chapter 7** — Apply core algorithms: community detection, centrality measures, and dynamics

For immediate exploration:

Browse the `examples/` directory in the repository for 50+ working examples covering various use cases.

Summary

This chapter covered:

1. **Quick install** — `pip install py3plex`
2. **Hello World** — Create a multilayer network in 30 seconds
3. **Basic analysis** — Statistics, visualization, and DSL queries
4. **Installation options** — Development, Docker, optional dependencies
5. **Key concepts** — Node-layer pairs, NetworkX compatibility, input formats

With py3plex installed, you're ready to analyze multilayer networks. The next chapter covers loading real data and representing complex multilayer structures.

1.7.5 Data Loading and Representation

This chapter covers how to load multilayer networks from various data sources, choose appropriate formats, and represent complex network structures correctly.

DSL Tip: Validate Data After Loading

Use DSL to quickly validate loaded data:

```
from py3plex.dsl import Q, L

# Check layer sizes
for layer in network.get_layers():
    count = Q.nodes().from_layers(L[layer]).execute(network).count
    print(f"{layer}: {count} nodes")

# Find potential data issues
```

```
isolated = Q.nodes().where(degree=0).execute(network)
if isolated.count > 0:
    print(f"Warning: {isolated.count} isolated nodes")

# Verify high-degree nodes make sense
hubs = (
    Q.nodes()
     .where(degree__gt=20)
     .compute("degree")
     .execute(network)
)
print(f"Hubs (degree > 20): {hubs.count}")
```

Quick validation catches data issues early!

Data Loading Basics

Multiple Input Methods

Py3plex supports several ways to create multilayer networks:

1. Direct edge addition (for programmatic construction):

```
from py3plex.core import multinet

network = multinet.multi_layer_network()
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Bob', 'friends', 'Carol', 'friends', 1],
], input_type="list")
```

2. File loading (for external data):

```
# Load from file
network = multinet.multi_layer_network()
network.load_network("data.edgelist", input_type="edgelist")
```

3. From NetworkX (for integration):

```
import networkx as nx

G = nx.karate_club_graph()
network = multinet.multi_layer_network()
network.load_network_from_networkx(G, layer_name='social')
```

Edgelist Format

The **edgelist** format is the simplest and most common:

```
# File: network.edgelist
Alice friends Bob friends 1.0
Bob friends Carol friends 1.0
Alice colleagues Bob colleagues 0.8
```

Format: `source source_layer target target_layer weight`

Loading:

```
network = multinet.multi_layer_network()
network.load_network("network.edgelist", input_type="edgelist")
```

Best practices:

- Use consistent node identifiers across layers
- Separate fields with spaces or tabs
- Include weights (default to 1.0 if omitted)
- Comment lines start with #

Dictionary Format

For programmatic construction, the dictionary format is most explicit:

```
network.add_edges([
    {
        'source': 'Alice',
        'source_type': 'friends',
        'target': 'Bob',
        'target_type': 'friends',
        'weight': 1.0,
        'timestamp': '2023-01-15'  # Optional attributes
    },
    {
        'source': 'Bob',
        'source_type': 'colleagues',
        'target': 'Carol',
        'target_type': 'colleagues',
        'weight': 0.8
    }
])
```

Advantages:

- Self-documenting (field names explicit)
- Supports arbitrary edge attributes
- Type-safe (Python dicts)
- Pythonic and readable

Modern I/O System

Py3plex provides a modern I/O system (`py3plex.io`) for high-performance serialization:

Supported Formats

Format	Extension	Best For	Trade-offs
JSON	`.json`	Human-readable, small networks	Slower, larger files
JSONL	`.jsonl`	Streaming, large networks	Not human-friendly
CSV	`.csv`	Spreadsheet tools, manual editing	Limited attributes
Arrow	`.arrow`	High performance, large networks	Requires pyarrow
Parquet	`.parquet`	Storage, compression, archiving	Requires pyarrow

Reading and Writing

```
from py3plex.io import read, write, MultiLayerGraph

# Write (auto-detects format from extension)
write(network, 'network.json')
write(network, 'network.arrow')
write(network, 'network.parquet')

# Read (auto-detects format)
graph = read('network.json')
graph = read('network.arrow')

# Specify format explicitly
write(network, 'myfile.dat', format='json')
graph = read('myfile.dat', format='parquet')
```

Apache Arrow Format (Recommended for Large Networks)

Arrow provides 2-3x faster I/O and better compression:

```
# Install Arrow support
pip install 'py3plex[arrow]'
```

```
from py3plex.io import read, write

# Feather format (fast, uncompressed)
write(graph, 'network.arrow')
graph = read('network.arrow')

# Parquet format (compressed, archival)
write(graph, 'network.parquet')
graph = read('network.parquet')
```

Performance comparison (1000 nodes, 5000 edges):

Format	Write Time	Read Time	File Size
Arrow	0.016s	0.008s	0.46 MB
Parquet	0.020s	0.010s	0.35 MB
JSON	0.046s	0.030s	1.09 MB

When to use Arrow:

- Large networks (>10k nodes)
- Performance-critical pipelines
- Interoperability with pandas, Spark, DuckDB
- Production data workflows

When to use JSON:

- Small networks (<1k nodes)
- Human readability required
- Debugging and manual editing
- Maximum compatibility

CSV with Sidecars

CSV format supports optional sidecar files for attributes:

```
# Write with sidecars
write(graph, 'edges.csv', write_sidecars=True)
# Creates: edges.csv, nodes.csv, layers.csv

# Read with sidecars
graph = read('edges.csv',
             nodes_file='nodes.csv',
             layers_file='layers.csv')
```

Data Validation and Best Practices

Always Verify After Loading

After loading external data, always verify structure:

```
network.load_network("data.edgelist", input_type="edgelist")

# Verify structure
print(f"Layers: {network.get_layers()}")
print(f"Nodes per layer: {network.get_number_of_nodes_per_layer()}")

# Check for unexpected layers
expected_layers = {'friends', 'colleagues'}
actual_layers = set(network.get_layers())
if actual_layers != expected_layers:
    print(f"Warning: unexpected layers {actual_layers - expected_layers}")
```

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```
# Basic statistics
network.basic_stats()
```

Consistent Node Identifiers

Rule: The same entity must have the same ID across all layers.

Good:

```
# Alice has the same ID in both layers
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Alice', 'colleagues', 'Carol', 'colleagues', 1],
], input_type="list")
```

Bad:

```
# Alice's ID differs across layers-breaks multiplex structure
network.add_edges([
    ['alice_social', 'friends', 'Bob', 'friends', 1],
    ['Alice_Work', 'colleagues', 'Carol', 'colleagues', 1],
], input_type="list")
```

Naming Conventions

Layer names:

- Use descriptive names: `'coauthor'`, `'twitter'`, not `'layer1'`, `'l2'`
- Lowercase with underscores: `'social_media'`, not `'SocialMedia'`
- Avoid special characters

Node IDs:

- Use stable, meaningful identifiers (user IDs, DOIs, names)
- Avoid auto-generated IDs that change across runs
- Preserve external IDs when loading from databases

Metadata and Attributes

Store metadata as node/edge attributes:

```
# Add node attributes
network.core_network.nodes[('Alice', 'friends')]['age'] = 30
network.core_network.nodes[('Alice', 'friends')]['country'] = 'USA'

# Add edge attributes
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1, {'type': 'close', 'since': 2018}]
], input_type="list")
```

Recommended practices:

- Use consistent attribute names across layers

- Store timestamps as ISO format strings: `'2023-01-15T10:30:00Z'`
- Use numeric types for attributes that will be analyzed
- Document attribute semantics in code comments

Representing Different Network Types

Multiplex Networks (Same Nodes Across Layers)

For networks where the same entities appear in all layers:

```
network = multinet.multi_layer_network(network_type='multiplex')

# Load data
network.load_network('social.edges', input_type='multiplex_edges')

# Coupling edges are automatically created
# (Alice, friends) <-> (Alice, colleagues)

# Get explicit edges only (exclude coupling)
explicit_edges = list(network.get_edges(multiplex_edges=False))
```

When to use:

- Same people across multiple social platforms
- Same cities in transportation modes (air, rail, road)
- Same proteins in different tissue types

Heterogeneous Networks (Different Node Types)

For networks with different entity types per layer:

```
network = multinet.multi_layer_network(network_type='multilayer')

# Different node types
network.add_edges([
    ['Alice', 'authors', 'Paper1', 'papers', 1],
    ['Bob', 'authors', 'Paper1', 'papers', 1],
    ['Paper1', 'papers', 'ICML', 'venues', 1],
], input_type="list")
```

When to use:

- Academic networks (authors, papers, venues)
- E-commerce (users, products, sellers)
- Biomedical (drugs, diseases, targets)

Temporal Networks (Time-Sliced Layers)

For networks that evolve over time:

```
network = multinet.multi_layer_network()
```

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```
# Time windows as layers
network.add_edges([
    ['Alice', '2020', 'Bob', '2020', 1],
    ['Alice', '2021', 'Bob', '2021', 1],
    ['Bob', '2021', 'Carol', '2021', 1],
], input_type="list")
```

Best practices:

- Use consistent time granularity (years, months, days)
- Layer names should be sortable: `'2020-01'`, `'2020-02'`, ...
- Add temporal edges connecting adjacent time slices if modeling evolution

Converting Between Formats

NetworkX to Py3plex

```
import networkx as nx
from py3plex.core import multinet

# Create NetworkX graph
G = nx.karate_club_graph()

# Convert to py3plex
network = multinet.multi_layer_network()
network.load_network_from_networkx(G, layer_name='social')
```

Py3plex to NetworkX

```
# Extract single layer as NetworkX graph
friends_layer = network.get_layer_subgraph('friends')

# Or access the full graph directly
full_graph = network.core_network  # MultiGraph or MultiDiGraph
```

Pandas DataFrame to Py3plex

```
import pandas as pd

# Load edges from DataFrame
df = pd.read_csv('edges.csv')
# Columns: source, source_layer, target, target_layer, weight

# Convert to list of lists
edges = df[['source', 'source_layer', 'target', 'target_layer', 'weight']].values.
↪tolist()

network = multinet.multi_layer_network()
network.add_edges(edges, input_type="list")
```

Common Data Loading Patterns

Pattern 1: Load from Multiple Files

```
network = multinet.multi_layer_network()

# Load different layers from different files
for layer_name, filename in [('friends', 'friends.txt'),
                             ('colleagues', 'colleagues.txt')]:
    with open(filename) as f:
        for line in f:
            source, target, weight = line.strip().split()
            network.add_edges([
                [source, layer_name, target, layer_name, float(weight)]
            ], input_type="list")
```

Pattern 2: Load from Database

```
import sqlite3

conn = sqlite3.connect('network.db')
cursor = conn.execute(
    "SELECT source, source_layer, target, target_layer, weight FROM edges"
)

network = multinet.multi_layer_network()
edges = [[row[0], row[1], row[2], row[3], row[4]] for row in cursor]
network.add_edges(edges, input_type="list")
```

Pattern 3: Incremental Loading (Large Networks)

```
network = multinet.multi_layer_network()

# Load in batches to manage memory
BATCH_SIZE = 10000

with open('large_network.edgelist') as f:
    batch = []
    for line in f:
        parts = line.strip().split()
        batch.append(parts)

        if len(batch) >= BATCH_SIZE:
            network.add_edges(batch, input_type="list")
            batch = []

    # Add remaining edges
    if batch:
        network.add_edges(batch, input_type="list")
```

Troubleshooting

Issue: Duplicate Edges

Symptom: More edges than expected after loading.

Solution: Check for duplicate entries in source data:

```
# Count edge multiplicity
edge_counts = {}
for edge in network.get_edges():
    edge_counts[edge] = edge_counts.get(edge, 0) + 1

duplicates = {e: c for e, c in edge_counts.items() if c > 1}
if duplicates:
    print(f"Found {len(duplicates)} duplicate edges")
```

Issue: Unexpected Layers

Symptom: Layers you didn't expect appear in the network.

Solution: Verify layer names in source data:

```
expected = {'friends', 'colleagues'}
actual = set(network.get_layers())

if actual != expected:
    print(f"Unexpected layers: {actual - expected}")
    print(f"Missing layers: {expected - actual}")
```

Issue: Node ID Mismatches

Symptom: Same entity appears as different nodes across layers.

Solution: Normalize node IDs before loading:

```
def normalize_id(node_id):
    """Normalize node identifiers."""
    return str(node_id).strip().lower()

# Apply during loading
edges = [[normalize_id(s), sl, normalize_id(t), tl, w]
         for s, sl, t, tl, w in raw_edges]
network.add_edges(edges, input_type="list")
```

Summary

This chapter covered:

1. **Loading methods** — Direct addition, files, NetworkX import
2. **Data formats** — Edgelist, dictionary, JSON, CSV, Arrow/Parquet
3. **Best practices** — Consistent IDs, validation, metadata storage
4. **Network types** — Multiplex, heterogeneous, temporal
5. **Common patterns** — Multi-file loading, databases, incremental loading

Key recommendations:

- Use **Arrow/Parquet** for large networks (>10k nodes)
- Use **JSON** for small networks and debugging
- **Always verify** structure after loading
- Use **consistent node IDs** across layers
- Store **metadata as attributes** for richer analysis

The next chapter covers visualization techniques for exploring multilayer network structure.

Further Reading

- Arrow format specification: https://arrow.apache.org/docs/
- NetworkX I/O reference: https://networkx.org/documentation/stable/reference/readwrite/
- Chapter 7 (Core Algorithms) for analysis after loading

1.7.6 Visualization and Exploration

Multilayer networks present unique visualization challenges: how do you clearly show multiple relationship types while preserving the overall structure? This chapter covers py3plex's visualization capabilities, from quick exploratory plots to publication-ready figures.

Overview

Visualizing multilayer networks requires balancing several concerns:

- **Clarity:** Individual layers must be distinguishable
- **Structure:** Cross-layer connections should be visible
- **Scale:** Techniques must work for networks of varying sizes (10 to 10,000+ nodes)
- **Purpose:** Exploratory plots need different settings than publication figures

py3plex provides three main visualization approaches:

1. **Static multilayer plots** — Show all layers simultaneously with customizable layouts
2. **Matrix visualizations** — Display the supra-adjacency matrix structure
3. **Layer-specific views** — Examine individual layers in detail

Quick Start

Basic Multilayer Visualization

```
from py3plex.core import multinet
from py3plex.visualization.multilayer import draw_multilayer_default

# Load network
network = multinet.multi_layer_network()
network.load_network("network.csv", input_type="multiedgelist")

# Visualize with defaults
draw_multilayer_default(
    network.get_layers(),
```

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```
    display=True,
    labels=True
)
```

This produces a circular layout with each layer shown in a different color.

Preset Visualization Modes

Py3plex provides three preset modes optimized for different network scales.

Minimal Mode (Large Networks >1000 nodes)

Optimized for networks where detail isn't critical and performance matters:

```
# Large network preset
draw_multilayer_default(
    network.get_layers(),
    node_size=5,              # Small nodes
    labels=False,             # No labels (too cluttered)
    edge_size=0.5,            # Thin edges
    alphalevel=0.3,           # Transparent edges
    background_shape="circle",
    display=True,
    remove_isolated_nodes=True
)
```

Use cases: Large social networks, overview visualizations, pattern detection at macro level.

Advantages: Fast rendering, reduced clutter, shows overall structure.

Balanced Mode (Medium Networks 100-1000 nodes)

Default settings that work well for most networks:

```
# Balanced preset (this is the default)
draw_multilayer_default(
    network.get_layers(),
    node_size=10,             # Medium nodes
    labels=True,              # Show layer labels
    edge_size=1.0,            # Normal edges
    alphalevel=0.13,          # Semi-transparent
    background_shape="circle",
    networks_color="rainbow",
    display=True
)
```

Use cases: Most research networks, exploratory analysis, publication figures with moderate detail.

Advantages: Good readability, reasonable performance, suitable for most publications.

Dense Mode (Small Networks <100 nodes)

Maximum detail for small networks where every node matters:

```
# Dense preset for detailed examination
draw_multilayer_default(
    network.get_layers(),
    node_size=20,              # Large nodes
    labels=True,
    node_labels=True,          # Show node IDs
    node_font_size=10,
    scale_by_size=True,        # Scale by degree
    edge_size=2.0,             # Thick edges
    alphalevel=0.8,            # Mostly opaque
    background_shape="rectangle",
    display=True
)
```

Use cases: Small case studies, detailed node-level analysis, high-quality publication figures.

Advantages: Maximum detail, clear node identification, professional appearance.

Layout Options

Circular Layout (Default)

Arranges layers in a circle, good for showing inter-layer connections:

```
draw_multilayer_default(
    network.get_layers(),
    background_shape="circle",
    rectanglex=1.0,  # Circle radius
    rectangley=1.0
)
```

Best for: 2-10 layers, symmetric interactions, publication figures.

Rectangular Layout

Arranges layers in a grid, good for many layers or hierarchical structures:

```
draw_multilayer_default(
    network.get_layers(),
    background_shape="rectangle",
    rectanglex=2.0,  # Width
    rectangley=1.0   # Height
)
```

Best for: Many layers (>10), hierarchical networks, time-series data, wide figures.

Auto-Scaling Features

Node Sizing by Degree

Automatically scale node sizes by their degree:

```
draw_multilayer_default(
    network.get_layers(),
    scale_by_size=True,    # Enable auto-scaling
    node_size=10           # Base size
)
```

Hub nodes become larger, making them easy to identify.

Color Assignment

Py3plex uses colorblind-safe palettes by default:

```
# Automatic rainbow colors
draw_multilayer_default(
    network.get_layers(),
    networks_color="rainbow"
)

# Custom palette
draw_multilayer_default(
    network.get_layers(),
    networks_color=["#FF6B6B", "#4ECDC4", "#45B7D1"]
)
```

Available palettes (from `py3plex.config`):

- `colorblind_safe`: 8 colors safe for colorblind viewers
- `wong`: 7-color Wong palette (scientifically validated)
- `tol_bright`: 7 bright colors

Matrix Visualizations

Supra-Adjacency Matrix

Visualize the full supra-adjacency matrix structure:

```
# Display matrix view
network.visualize_matrix({"display": True})
```

This shows the block structure: diagonal blocks are intra-layer edges, off-diagonal blocks are inter-layer edges.

Use cases: Understanding network structure, debugging layer connections, small networks only (<500 nodes).

Exploration Workflows

Layer Views

Examine individual layers:

```
# Extract and visualize a single layer
layer_subgraph = network.get_layer("social")

import networkx as nx
import matplotlib.pyplot as plt
```

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```
nx.draw(layer_subgraph, with_labels=True)
plt.show()
```

Cross-Layer Pattern Analysis

Use the DSL to filter and visualize subsets:

```
from py3plex.dsl import Q, L

# Find high-degree nodes across specific layers
result = (
    Q.nodes()
     .from_layers(L["social"] + L["professional"])
     .where(degree__gt=10)
     .execute(network)
)

# Extract subgraph for visualization
high_degree_nodes = list(result.node_ids)
subgraph = network.core_network.subgraph(high_degree_nodes)
```

Node Neighborhoods

Extract ego networks (neighborhood around a specific node):

```
import networkx as nx

# Get 2-hop neighborhood around 'Alice'
alice_ego = nx.ego_graph(network.core_network, ('Alice', 'social'), radius=2)

# Visualize
nx.draw(alice_ego, with_labels=True)
```

Performance Considerations

Visualization performance depends heavily on network size:

- **<100 nodes:** All visualization modes work well
- **100-1000 nodes:** Use balanced or minimal mode; avoid node labels
- **1000-5000 nodes:** Use minimal mode; rendering may be slow
- **>5000 nodes:** Static visualizations become impractical; use matrix view or export to specialized tools

Tips for large networks:

1. Use `remove_isolated_nodes=True` to reduce clutter
2. Set `alphalevel=0.1` or lower for edge transparency
3. Disable labels: `labels=False, node_labels=False`
4. Consider sampling: visualize only high-degree nodes or a random subset

Summary

Key visualization capabilities in py3plex:

- **Three preset modes** (minimal, balanced, dense) for different network scales
- **Flexible layouts** (circular, rectangular) for different structures
- **Auto-scaling** features adapt to network properties
- **Matrix visualizations** show supra-adjacency structure
- **Layer-specific views** enable detailed exploration

Choose visualization settings based on your network size and purpose. For quick exploration, use defaults. For publications, use dense mode with custom colors and layouts.

Next chapter: Core algorithms for multilayer analysis (community detection, centrality, dynamics)

1.7.7 Core Algorithms: Communities, Centrality, Dynamics

This chapter covers py3plex's three major algorithm families for multilayer network analysis: community detection, centrality measures, and dynamics/spreading processes.

DSL Tip: Streamline Algorithm Workflows

The DSL simplifies many algorithmic workflows:

```
from py3plex.dsl import Q, L

# Quick centrality analysis
top_central = (
    Q.nodes()
     .from_layers(L["social"])
     .compute("betweenness_centrality", "pagerank", "degree")
     .order_by("-betweenness_centrality")
     .limit(20)
     .execute(network)
)

# Multi-layer comparison
for layer in ["social", "work"]:
    result = (
        Q.nodes()
         .from_layers(L[layer])
         .compute("betweenness_centrality")
         .execute(network)
    )
    print(f"{layer}: avg BC = {sum(result.measures['betweenness_centrality'].
↪values()) / result.count:.4f}")
```

See Chapter 8 for comprehensive DSL coverage!

Overview

[Introduce three major algorithm families for multilayer analysis]

Community Detection

Multilayer Modularity

[Mathematical definition and implementation]

```
from py3plex.algorithms.community_detection import multilayer_louvain

# Detect communities
communities = multilayer_louvain.best_partition(network.core_network)
```

Algorithms Available

[Louvain, Infomap, Label Propagation]

Complexity: O(n log n) for most algorithms

Choosing a Community Detection Method

[Guidelines based on network size, structure, and goals]

Centrality Measures

Multilayer PageRank

[Random walk-based centrality]

Degree Centrality

[Intra-layer vs inter-layer degree]

Betweenness Centrality

[Path-based importance]

Explainable Centrality

[Breaking down centrality scores by layer]

```
from py3plex.algorithms.centrality.explain import explain_node_centrality

# Get layer-wise breakdown
explanation = explain_node_centrality(network, 'Alice', measure='degree')
```

Dynamics and Processes

py3plex provides comprehensive support for simulating dynamical processes on multilayer networks, including epidemic models and random walks.

For complete documentation, see the user guide section on multilayer dynamics and the SIR epidemic simulator documentation.

Random Walks

Random walks simulate exploration and diffusion on networks:

```
from py3plex.dynamics import RandomWalkDynamics

# Create random walk
walk = RandomWalkDynamics(network, start_node=0, lazy_probability=0.1)
walk.set_seed(42)
results = walk.run(steps=1000)

# Analyze trajectory
trajectory = results.get_measure("trajectory")
```

Random walks are fundamental to PageRank, community detection, and network embeddings (Node2Vec, DeepWalk).

Epidemic Models

Simulate disease spread and information diffusion with SIR and SIS models:

```
from py3plex.dynamics import SIRDynamics, SISDynamics

# SIR: Susceptible-Infected-Recovered (with immunity)
sir = SIRDynamics(network, beta=0.3, gamma=0.1, initial_infected=0.1)
sir.set_seed(42)
results = sir.run(steps=100)

# Extract measures
prevalence = results.get_measure("prevalence")
state_counts = results.get_measure("state_counts")

print(f"Peak prevalence: {prevalence.max():.2%}")
print(f"Final recovered: {state_counts['R'][-1]}")
```

Key parameters:

- **beta ()**: Transmission rate (infection probability per contact)
- **gamma ()**: Recovery rate (recovery probability per time step)
- **R_0 = / × ⟨k⟩**: Basic reproduction number (epidemic threshold at R_0=1)

Models:

- **SIR** — Epidemic with immunity (measles, chickenpox). Always dies out eventually.
- **SIS** — Endemic disease without immunity (common cold, malware). Reaches stable equilibrium.

Multilayer Dynamics

Infection spreads through multiple interaction channels:

```
# Two-layer network: physical + digital contacts
network = multinet.multi_layer_network(directed=False)

# Add physical layer (local connections)
# Add digital layer (global connections)
```

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```
# Run dynamics
sir = SIRDynamics(network, beta=0.3, gamma=0.1)
results = sir.run(steps=100)
```

The multilayer structure allows simultaneous spread through different contexts, often leading to faster and more extensive diffusion than single-layer networks.

Algorithm Complexity and Scaling

[Performance characteristics, memory requirements, when to use each]

Summary

[Recap of algorithm families and when to use each]

Next chapter: Introduction to the py3plex DSL

Source files to integrate: - docfiles/user_guide/community_detection.rst - docfiles/user_guide/statistics.rst - docfiles/tutorials/multilayer_centrality.rst - docfiles/tutorials/explainable_centrality.rst - docfiles/sir_epidemic_simulator.rst - DYNAMICS_IMPLEMENTATION.md - docfiles/multilayer_centrality_matrix_functions.rst - docfiles/ricci_curvature.rst

1.7.8 Introduction to the Py3plex DSL

This chapter introduces the py3plex Domain-Specific Language (DSL), a SQL-like query language for expressing multilayer network analyses concisely. The DSL is a **major first-class feature** that sets py3plex apart from other network libraries.

DSL at a Glance

The DSL provides intuitive SQL-like syntax for network queries:

```
from py3plex.dsl import Q, L

# Simple: Find high-degree nodes
result = Q.nodes().where(degree__gt=5).execute(network)

# Powerful: Multi-layer analysis with export
result = (
    Q.nodes()
     .from_layers(L["social"] + L["work"])
     .where(degree__gt=3)
     .compute("betweenness_centrality", "pagerank")
     .order_by("-betweenness_centrality")
     .limit(20)
     .export_csv("top_influencers.csv")
     .execute(network)
)
```

Express complex analyses in a few readable lines!

Why a DSL for Networks?

Traditional network analysis requires writing explicit loops and conditionals:

```
# Traditional approach: verbose and error-prone
high_degree_nodes = []
for node in network.get_nodes():
    if node[1] == 'social':  # Check layer
        degree = network.core_network.degree(node)
        if degree > 5:
            high_degree_nodes.append(node)

# Compute centrality for filtered nodes
centralities = {}
for node in high_degree_nodes:
    centralities[node] = nx.betweenness_centrality(G)[node]
```

The DSL provides a declarative alternative:

```
# DSL approach: clear and concise
from py3plex.dsl import execute_query

result = execute_query(network,
    'SELECT nodes WHERE layer="social" AND degree > 5 '
    'COMPUTE betweenness_centrality'
)
```

Benefits:

1. **Declarative** — State *what* you want, not *how* to get it
2. **Readable** — SQL-like syntax is familiar and self-documenting
3. **Composable** — Build complex queries from simple building blocks
4. **Maintainable** — Less code means fewer bugs

DSL Versions

Py3plex provides two DSL interfaces:

String DSL (v1)
SQL-like queries as strings. Good for simple queries and REPL exploration.

```
result = execute_query(network, 'SELECT nodes WHERE degree > 5')
```

Builder API (v2, recommended)
Python-native chainable API with type hints and autocompletion.

```
from py3plex.dsl import Q, L

result = (
    Q.nodes()
     .from_layers(L["social"])
     .where(degree__gt=5)
     .execute(network)
)
```

This chapter focuses on the **Builder API (v2)** as it provides a better development experience. The string DSL is covered briefly for reference.

Basic Query Structure

A typical DSL query has four parts:

1. **Target** — What to select (nodes or edges)
2. **Layer filtering** — Which layers to include (optional)
3. **Conditions** — Filter criteria (optional)
4. **Measures** — What to compute (optional)

Example:

```
from py3plex.dsl import Q, L

result = (
    Q.nodes()                          # 1. Select nodes
     .from_layers(L["social"])         # 2. Filter to social layer
     .where(degree__gt=5)              # 3. Filter by degree > 5
     .compute("betweenness_centrality")  # 4. Compute centrality
     .execute(network)                 # Execute query
)
```

Your First Query

Let's build a simple query step-by-step.

Setup:

```
from py3plex.core import multinet
from py3plex.dsl import Q, L

# Create network
network = multinet.multi_layer_network()
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Bob', 'friends', 'Carol', 'friends', 1],
    ['Carol', 'friends', 'Dave', 'friends', 1],
    ['Alice', 'colleagues', 'Bob', 'colleagues', 1],
    ['Bob', 'colleagues', 'Eve', 'colleagues', 1],
], input_type="list")
```

Query 1: Select all nodes

```
result = Q.nodes().execute(network)

print(f"Count: {result.count}")
# Output: Count: 7

for node in result:
    print(node)
# Output: ('Alice', 'friends'), ('Bob', 'friends'), ...
```

Query 2: Filter by layer

```
result = Q.nodes().from_layers(L["friends"]).execute(network)

print(f"Nodes in friends layer: {result.count}")
# Output: Nodes in friends layer: 4
```

Query 3: Filter by degree

```
result = Q.nodes().where(degree__gt=1).execute(network)

print(f"High-degree nodes: {result.count}")
# Output: High-degree nodes: 2 (Bob in both layers)
```

Query 4: Compute centrality

```
result = (
    Q.nodes()
     .from_layers(L["friends"])
     .compute("betweenness_centrality")
     .execute(network)
)

for node, centrality in result.measures['betweenness_centrality'].items():
    print(f"{node}: {centrality:.3f}")
# Output: ('Bob', 'friends'): 1.000, others: 0.000
```

Layer Filtering

The DSL provides powerful layer algebra operations.

DSL Layer Algebra

Layer algebra lets you combine layers with set operations:

```
from py3plex.dsl import Q, L

# Union: nodes in social OR work
Q.nodes().from_layers(L["social"] + L["work"])

# Difference: nodes in social but NOT bots
Q.nodes().from_layers(L["social"] - L["bots"])

# Intersection: nodes in both social AND work
Q.nodes().from_layers(L["social"] & L["work"])

# Complex: (social OR work) - bots
Q.nodes().from_layers(L["social"] + L["work"] - L["bots"])
```

This makes multilayer queries intuitive and expressive!

Simple Layer Selection

```
# Single layer
Q.nodes().from_layers(L["social"])

# Multiple layers (union)
Q.nodes().from_layers(L["social"] + L["work"])
```

Layer Algebra

Union (OR):

```
# Nodes in social OR work layers
Q.nodes().from_layers(L["social"] + L["work"])
```

Difference (NOT):

```
# Nodes in social but NOT in bots layer
Q.nodes().from_layers(L["social"] - L["bots"])
```

Intersection (AND):

```
# Nodes present in both social AND work
Q.nodes().from_layers(L["social"] & L["work"])
```

All layers:

```
# Don't specify from_layers() to query all layers
Q.nodes().where(degree__gt=5)
```

Filtering Conditions

The `where()` method accepts filtering predicates.

Degree Filters

```
# Greater than
Q.nodes().where(degree__gt=5)

# Greater than or equal
Q.nodes().where(degree__gte=3)

# Less than
Q.nodes().where(degree__lt=10)

# Less than or equal
Q.nodes().where(degree__lte=5)

# Equal
Q.nodes().where(degree=2)
```

Layer Filters

```
# Specific layer
Q.nodes().where(layer="social")

# Exclude layer
Q.nodes().where(layer__ne="bots")
```

Multiple Conditions (AND)

```
# Combine conditions
Q.nodes().where(layer="social", degree__gt=3)

# Equivalent to: layer="social" AND degree > 3
```

Computing Measures

The `compute()` method calculates network measures.

Single Measure

```
result = (
    Q.nodes()
     .compute("degree")
     .execute(network)
)

# Access results
for node, degree in result.measures['degree'].items():
    print(f"{node}: {degree}")
```

Multiple Measures

```
result = (
    Q.nodes()
     .compute("degree", "betweenness_centrality", "closeness_centrality")
     .execute(network)
)

# Access each measure
degrees = result.measures['degree']
betweenness = result.measures['betweenness_centrality']
closeness = result.measures['closeness_centrality']
```

Available Measures

Measure	Description
`degree`	Node degree (number of neighbors)
`degree_centrality`	Normalized degree centrality
`betweenness_centrality`	Betweenness centrality (path-based importance)
`closeness_centrality`	Closeness centrality (average distance to others)
`eigenvector_centrality`	Eigenvector centrality (importance of neighbors)
`pagerank`	PageRank score
`clustering`	Clustering coefficient

Ordering and Limiting

Order By

```
# Order by degree (ascending)
Q.nodes().compute("degree").order_by("degree")

# Order by degree (descending)
Q.nodes().compute("degree").order_by("-degree")

# Order by multiple keys
Q.nodes().compute("degree").order_by("-degree", "node_id")
```

Limit

```
# Get top 10 by degree
result = (
    Q.nodes()
     .compute("degree")
     .order_by("-degree")
     .limit(10)
     .execute(network)
)
```

Working with Results

The `QueryResult` object provides multiple ways to access results.

Iteration

```
result = Q.nodes().execute(network)

for node in result:
    print(node)  # ('Alice', 'friends'), ...
```

Count

```
result = Q.nodes().where(degree__gt=5).execute(network)
print(f"Found {result.count} nodes")
```

Measures

```
result = Q.nodes().compute("degree").execute(network)

# Access as dict
degrees = result.measures['degree']
for node, degree in degrees.items():
    print(f"{node}: {degree}")
```

To Pandas

```
# Convert to DataFrame
df = result.to_pandas()
print(df.head())
```

String DSL Syntax (Reference)

For completeness, here's the string DSL syntax:

Basic queries:

```
# Select nodes
execute_query(network, 'SELECT nodes')

# Filter by layer
execute_query(network, 'SELECT nodes WHERE layer="social"')

# Filter by degree
execute_query(network, 'SELECT nodes WHERE degree > 5')

# Multiple conditions
execute_query(network, 'SELECT nodes WHERE layer="social" AND degree > 3')

# Compute measures
execute_query(network, 'SELECT nodes COMPUTE betweenness_centrality')
```

Operators:

- = — Equal
- != — Not equal
- > — Greater than
- < — Less than
- >= — Greater or equal
- <= — Less or equal
- `AND` — Logical AND

- `OR` — Logical OR
- `NOT` — Logical NOT

Example: Complete Workflow

Here's a complete example demonstrating DSL capabilities:

```
from py3plex.core import multinet
from py3plex.dsl import Q, L

# Create network
network = multinet.multi_layer_network()
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Bob', 'friends', 'Carol', 'friends', 1],
    ['Carol', 'friends', 'Dave', 'friends', 1],
    ['Alice', 'colleagues', 'Bob', 'colleagues', 1],
    ['Bob', 'colleagues', 'Eve', 'colleagues', 1],
    ['Eve', 'colleagues', 'Frank', 'colleagues', 1],
], input_type="list")

# Query 1: Find high-degree nodes across all layers
result = (
    Q.nodes()
     .where(degree__gte=2)
     .compute("degree", "betweenness_centrality")
     .order_by("-betweenness_centrality")
     .execute(network)
)

print("High-degree nodes:")
for node in result:
    degree = result.measures['degree'][node]
    bc = result.measures['betweenness_centrality'][node]
    print(f"  {node}: degree={degree}, BC={bc:.3f}")

# Query 2: Compare layers
for layer in ['friends', 'colleagues']:
    result = (
        Q.nodes()
         .from_layers(L[layer])
         .compute("degree")
         .execute(network)
    )
    avg_degree = sum(result.measures['degree'].values()) / result.count
    print(f"{layer} layer: {result.count} nodes, avg degree={avg_degree:.2f}")
```

Common Patterns

Pattern 1: Filter → Compute → Order → Limit

```
# Find top 10 central nodes in a specific layer
top_10 = (
```

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```
    Q.nodes()
      .from_layers(L["social"])
      .compute("betweenness_centrality")
      .order_by("-betweenness_centrality")
      .limit(10)
      .execute(network)
)
```

Pattern 2: Layer Comparison

```
# Compare degree distribution across layers
for layer_name in network.get_layers():
    result = (
        Q.nodes()
          .from_layers(L[layer_name])
          .compute("degree")
          .execute(network)
    )
    degrees = list(result.measures['degree'].values())
    print(f"{layer_name}: mean degree = {sum(degrees)/len(degrees):.2f}")
```

Pattern 3: Conditional Export

```
# Export high-degree nodes to CSV
(
    Q.nodes()
      .where(degree__gt=5)
      .compute("degree", "betweenness_centrality")
      .order_by("-degree")
      .export_csv("high_degree_nodes.csv")
      .execute(network)
)
```

Summary

This chapter introduced the py3plex DSL:

1. **Motivation** — Declarative queries replace explicit loops
2. **Builder API** — Python-native with type hints (recommended)
3. **Query structure** — Target, layers, conditions, measures
4. **Layer algebra** — Union, difference, intersection operations
5. **Results** — Iteration, measures, pandas export

Key takeaways:

- Use **Builder API** for type safety and autocompletion
- **Chain methods** to build complex queries incrementally
- **Layer algebra** enables powerful layer filtering
- **Compute measures** in a single call instead of manual loops

The next chapter covers the EXPLAIN mode for understanding query execution and performance.

Further Reading

- The Builder API and Explain Plans
- Advanced Queries and Workflows
- `examples/network_analysis/example_dsl_builder_api.py` — Complete examples

1.7.9 The Builder API and Explain Plans

The DSL Builder API provides a Pythonic, type-safe way to construct network queries using method chaining. This chapter covers the builder pattern, query execution plans, and advanced features like parameterization and query reuse.

Builder Pattern and Fluent API

The builder API uses method chaining to construct queries incrementally. Each method returns the query object, allowing you to chain operations naturally:

```
from py3plex.dsl import Q, L

# Basic query with chaining
result = (
    Q.nodes()                              # Start with nodes
     .from_layers(L["social"])             # Filter by layer
     .where(degree__gt=5)                  # Filter by degree
     .compute("betweenness_centrality") # Compute measure
     .order_by("-betweenness_centrality") # Sort descending
     .limit(10)                            # Take top 10
     .execute(network)                     # Execute query
)
```

Advantages over string DSL:

- **Type safety:** IDE autocomplete and type checking
- **Composability:** Build queries programmatically
- **Readability:** Self-documenting code
- **Error detection:** Catch mistakes before execution

Method Chaining

Build queries step-by-step:

```
# Start with a base query
q = Q.nodes()

# Add filters
q = q.where(layer="social")
q = q.where(degree__gt=3)

# Add computation
q = q.compute("degree", "clustering")
```

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```
# Sort and limit
q = q.order_by("-degree")
q = q.limit(20)

# Execute
result = q.execute(network)
```

Builder operators:

Operator	DSL Equivalent
`where(degree__gt=5)`	`WHERE degree > 5`
`where(degree__gte=5)`	`WHERE degree >= 5`
`where(degree__lt=5)`	`WHERE degree < 5`
`where(degree__lte=5)`	`WHERE degree <= 5`
`where(layer="social")`	`WHERE layer = "social"`
`where(layer__in=["a", "b"])`	`WHERE layer IN ("a", "b")`

Type Hints and IDE Support

The builder API is fully type-hinted for modern IDEs:

```
from py3plex.dsl import Q, L

# IDE shows available methods and parameters
result = (
    Q.nodes()           # Autocomplete suggests: where, from_layers, compute, ...
     .where(            # Autocomplete shows: degree__gt, degree__lt, layer, ...
         degree__gt=5   # Type hints ensure correct parameter types
     )
     .compute(          # Autocomplete lists available measures
         "degree",      # String literals with validation
         "betweenness_centrality"
     )
     .execute(network)  # Type checker ensures network is correct type
)
```

Benefits:

- Catch typos before runtime
- Discover available methods and measures
- Understand query structure through types
- Safer refactoring with IDE support

EXPLAIN Mode

Query Execution Plans

Get a query execution plan without actually running the query:

```python
from py3plex.dsl import Q

# Build a complex query
q = (
    Q.nodes()
     .from_layers(L["social"])
     .where(degree__gt=5)
     .compute("betweenness_centrality", "pagerank")
     .order_by("-betweenness_centrality")
     .limit(10)
)

# Get execution plan
plan = q.explain().execute(network)

# Inspect the plan
print(f"Query: {plan.query}")
print(f"Estimated nodes: {plan.estimated_node_count}")

for step in plan.steps:
    print(f"  {step.description}")
    print(f"    Complexity: {step.estimated_complexity}")
    print(f"    Estimated time: {step.estimated_time_ms}ms")

# Check for warnings
for warning in plan.warnings:
    print(f"   {warning}")
```

Example output:

```
Query: SELECT nodes FROM layers social WHERE degree > 5
       COMPUTE betweenness_centrality pagerank
       ORDER BY -betweenness_centrality LIMIT 10

Estimated nodes: ~150

  1. Filter by layer
    Complexity: O(n)
    Estimated time: 2ms

  2. Filter by degree
    Complexity: O(n)
    Estimated time: 1ms

  3. Compute betweenness_centrality
    Complexity: O(n^3)
    Estimated time: 850ms
     Betweenness centrality is expensive for large graphs

  4. Compute pagerank
    Complexity: O(n^2)
    Estimated time: 45ms
```

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```
  5. Sort results
    Complexity: O(n log n)
    Estimated time: 3ms

  6. Limit results
    Complexity: O(1)
    Estimated time: <1ms
```

Complexity Estimates

EXPLAIN mode provides complexity estimates for each operation:

- **O(1)** — Constant time (e.g., limit, basic metadata)
- **O(n)** — Linear in number of nodes (e.g., filtering by attribute)
- **O(m)** — Linear in number of edges (e.g., edge filtering)
- **O(n log n)** — Log-linear (e.g., sorting)
- **O(n^2)** — Quadratic (e.g., PageRank, all-pairs measures)
- **O(n^3)** — Cubic (e.g., betweenness centrality on large graphs)

Use EXPLAIN to:

1. **Identify bottlenecks** before executing expensive queries
2. **Compare alternative formulations** of the same query
3. **Estimate runtime** for large networks
4. **Optimize query order** (apply cheap filters first)

Optimization Tips

1. Apply filters before computing measures

```
#  Slow: Compute for all, then filter
Q.nodes().compute("betweenness_centrality").where(degree__gt=5)

#  Fast: Filter first, then compute
Q.nodes().where(degree__gt=5).compute("betweenness_centrality")
```

2. Use layer filtering early

```
#  Slow: Compute for all layers
Q.nodes().compute("degree").where(layer="social")

#  Fast: Filter to layer first
Q.nodes().from_layers(L["social"]).compute("degree")
```

3. Limit results when possible

```
# Top 10 is much faster than computing for all nodes
Q.nodes().compute("degree").order_by("-degree").limit(10)
```

4. Choose efficient measures

- **Cheap:** degree, clustering coefficient (local measures)
- **Moderate:** PageRank, eigenvector centrality (iterative)
- **Expensive:** betweenness centrality, closeness centrality (all-pairs)

Advanced Builder Features

Parameterized Queries

Use `Param` to create reusable query templates:

```
from py3plex.dsl import Q, Param

# Create a parameterized query template
influencer_query = (
    Q.nodes()
     .from_layers(L[Param.str("target_layer")])
     .where(degree__gt=Param.int("min_degree"))
     .compute("betweenness_centrality")
     .order_by("-betweenness_centrality")
     .limit(Param.int("top_n"))
)

# Execute with different parameters
social_influencers = influencer_query.execute(
    network,
    target_layer="social",
    min_degree=10,
    top_n=20
)

work_influencers = influencer_query.execute(
    network,
    target_layer="professional",
    min_degree=5,
    top_n=50
)
```

Parameter types:

- `Param.int("name")` — Integer parameter
- `Param.float("name")` — Float parameter
- `Param.str("name")` — String parameter
- `Param.ref("name")` — Untyped reference

Benefits:

- **Safety:** Type-checked at execution time
- **Reusability:** One query definition, many executions
- **Maintainability:** Parameters are self-documenting
- **Performance:** Query is parsed once, executed multiple times

Query Reuse

Build queries once and execute them multiple times:

```
# Define a query once
high_degree_analysis = (
    Q.nodes()
     .where(degree__gt=10)
     .compute("betweenness_centrality", "clustering")
)

# Execute on different networks
result1 = high_degree_analysis.execute(network1)
result2 = high_degree_analysis.execute(network2)
result3 = high_degree_analysis.execute(network3)

# Or execute with different parameters over time
for threshold in [5, 10, 15, 20]:
    q = Q.nodes().where(degree__gt=threshold).compute("degree")
    result = q.execute(network)
    print(f"Threshold {threshold}: {result.count} nodes")
```

Converting to DSL String

Convert a builder query back to string format:

```
q = (
    Q.nodes()
     .from_layers(L["social"])
     .where(degree__gt=5)
     .compute("degree")
     .limit(10)
)

# Get DSL string representation
dsl_string = q.to_dsl()
print(dsl_string)
# Output: SELECT nodes FROM layers social WHERE degree > 5
#         COMPUTE degree LIMIT 10
```

Use cases:

- **Logging:** Record queries in application logs
- **Debugging:** Inspect query structure
- **Serialization:** Save queries to files or databases
- **Auditing:** Track what analyses were run

Error Handling with Suggestions

The DSL provides helpful error messages with suggestions:

```
from py3plex.dsl import Q, UnknownMeasureError
```

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```
try:
    # Typo in measure name
    result = Q.nodes().compute("betweenes").execute(network)
except UnknownMeasureError as e:
    print(e)
    # Output: Unknown measure 'betweenes'. Did you mean 'betweenness_centrality'?
    #         Known measures: betweenness_centrality, closeness_centrality, ...
```

Summary

The DSL Builder API provides:

- **Pythonic interface:** Natural method chaining
- **Type safety:** IDE support and autocomplete
- **Query plans:** EXPLAIN mode for optimization
- **Parameterization:** Reusable query templates
- **Error handling:** Helpful "did you mean?" messages

Best practices:

1. Use builder API for programmatic queries
2. Run EXPLAIN before executing expensive queries
3. Filter early, compute late
4. Parameterize queries that you'll reuse
5. Use type hints and IDE support to catch errors

Next chapter: Advanced query patterns and workflow integration

1.7.10 Advanced Queries and Workflows

This chapter covers advanced DSL patterns including dynamics simulation, complex query patterns, and workflow integration.

DSL-Based Dynamics Simulation

The py3plex DSL includes a powerful framework for declarative dynamics simulation on multilayer networks. This section demonstrates how to use the dynamics DSL for epidemic modeling, random walks, and other dynamical processes.

Mathematical Formalism

SIS Model (Susceptible-Infected-Susceptible)

The SIS model tracks disease spread with possible reinfection:

- **States:** S (susceptible), I (infected)
- **Update rules:**
  - S $\rightarrow$ I with probability = 1 - (1 -)^(A · I)
  - I $\rightarrow$ S with probability
- **Parameters:**

  - : transmission probability per contact (0 1)
  - : recovery probability per time step (0 1)

The SIS model exhibits endemic equilibria when $R_0 = / \cdot {}_1(A) > 1$, where ${}_1(A)$ is the largest eigenvalue of the adjacency matrix.

SIR Model (Susceptible-Infected-Recovered)

The SIR model includes permanent immunity after recovery:

- **States:** S (susceptible), I (infected), R (recovered/removed)
- **Update rules:**
  - S $\rightarrow$ I with probability = 1 - (1 -)^(A · I)
  - I $\rightarrow$ R with probability
- **Parameters:**
  - : transmission probability per contact
  - : recovery probability per time step

The final outbreak size (attack rate) depends on the basic reproduction number $R_0 = / \cdot \langle k \rangle$, where $\langle k \rangle$ is the mean degree.

Random Walk

Random walk dynamics model diffusion processes:

- **State:** Current node position
- **Update rule:** Move to neighbor j with probability 1/degree(i), or stay at i with probability p_lazy (for lazy walks)
- **Parameters:**
  - teleport: probability of random teleportation (default: 0.05)

Basic Dynamics DSL Usage

The dynamics DSL uses a builder API similar to the query DSL:

```
from py3plex.core import multinet
from py3plex.dynamics import D, SIS
from py3plex.dsl import L

# Create network
network = multinet.multi_layer_network()
# ... add nodes and edges ...

# Define SIS simulation
sim = (
    D.process(SIS(beta=0.3, mu=0.1))  # Specify process and parameters
     .initial(infected=0.05)           # 5% initially infected
     .steps(100)                       # Run for 100 time steps
     .measure("prevalence", "incidence")  # Track measures
     .replicates(10)                   # Run 10 independent simulations
     .seed(42)                         # For reproducibility
)
```

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```
# Execute simulation
result = sim.run(network)

# Access results
print(f"Mean final prevalence: {result.data['prevalence'][:, -1].mean():.3f}")

# Convert to pandas for analysis
df_dict = result.to_pandas()
prevalence_df = df_dict['prevalence']
```

Key components:

1. **D.process()** — Specify the dynamical process (SIS, SIR, RandomWalk)
2. **.initial()** — Set initial conditions
3. **.steps()** — Number of time steps to simulate
4. **.measure()** — Metrics to track during simulation
5. **.replicates()** — Number of independent runs (for averaging)
6. **.seed()** — Random seed for reproducibility
7. **.run()** — Execute the simulation

Available Measures

Different processes support different measures:

Process	Measure	Description
SIS	prevalence	Fraction of infected nodes at each time step
SIS	incidence	Number of new infections at each time step
SIS	prevalence_by_layer	Prevalence tracked separately per layer
SIR	prevalence	Fraction of infected nodes
SIR	state_counts	Counts of nodes in each state (S, I, R)
RandomWalk	visit_frequency	Frequency of visits to each node
RandomWalk	state_counts	Current position over time

Multilayer Dynamics

The DSL supports multilayer networks with coupling between layers:

```
from py3plex.dsl import L

# Simulate on specific layers
sim = (
    D.process(SIS(beta=0.25, mu=0.08))
     .on_layers(L["offline"] + L["online"])  # Select layers
     .coupling(node_replicas="strong")       # Shared node states
     .initial(infected=0.1)
     .steps(120)
     .measure("prevalence", "prevalence_by_layer")
     .replicates(15)
```

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```
)

result = sim.run(multilayer_network)
```

Coupling modes:

- **"strong"** — Node states shared across all layers (default)
- **"independent"** — Each layer has independent node states
- **"weak"** — Partial coupling with cross-layer transmission rate

Integration with Query DSL

The dynamics DSL integrates seamlessly with the query DSL for targeted initial conditions:

```
from py3plex.dsl import Q

# Start infection at high-degree nodes (hubs)
sim = (
    D.process(SIS(beta=0.35, mu=0.12))
     .initial(
         infected=Q.nodes().where(degree__gte=5)  # Query for hubs
     )
     .steps(100)
     .measure("prevalence")
     .replicates(10)
)

result = sim.run(network)
```

This allows precise control over initial conditions based on network structure, centrality, or other properties.

Parameter Comparison Example

Compare dynamics under different parameters:

```
# Compare transmission rates
beta_values = [0.2, 0.3, 0.4, 0.5]
results = {}

for beta in beta_values:
    sim = (
        D.process(SIS(beta=beta, mu=0.1))
         .initial(infected=0.05)
         .steps(80)
         .measure("prevalence")
         .replicates(20)
         .seed(42)
    )
    results[beta] = sim.run(network)

# Analyze final prevalence
for beta, result in results.items():
```

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```
    mean_final = result.data['prevalence'][:, -1].mean()
    print(f"={beta:.1f}: final prevalence = {mean_final:.3f}")
```

Result Analysis

The `SimulationResult` object provides rich analysis capabilities:

```
# Get summary statistics
summary = result.summary()
print(summary)

# Plot time series with confidence intervals
import matplotlib.pyplot as plt
result.plot("prevalence")
plt.show()

# Export to pandas for custom analysis
df_dict = result.to_pandas()
prevalence_df = df_dict['prevalence']

# Compute mean trajectory
mean_trajectory = (
    prevalence_df
    .groupby('t')['value']
    .agg(['mean', 'std'])
)
```

Status: Dynamics DSL

The dynamics DSL API shown above is **experimental**. The core dynamics classes (`SIRDynamics`, `SISDynamics`, `RandomWalkDynamics`) are stable and production-ready. The builder API (`D.process()`) is under active development.

Complex Query Patterns

Beyond basic filtering and computation, the DSL supports advanced analysis patterns.

Parameterized Comparative Analysis

Systematic comparison across layers using parameterized queries:

```
from py3plex.dsl import Q, L, Param

# Define reusable parameterized query
hub_analysis = (
    Q.nodes()
     .from_layers(L[Param.str("layer")])
     .where(degree__gt=Param.int("threshold"))
     .compute("betweenness_centrality", "clustering")
     .order_by("-betweenness_centrality")
     .limit(Param.int("top_n"))
```

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```
)

# Execute for multiple layers
for layer in ["social", "work", "family"]:
    result = hub_analysis.execute(
        network,
        layer=layer,
        threshold=5,
        top_n=20
    )
    df = result.to_pandas()
    df.to_csv(f"{layer}_hubs.csv")
    print(f"{layer}: {result.count} hubs, max BC = {df['betweenness_centrality'].max():.
↪3f}")
```

Multi-Layer Statistical Summaries

Aggregate statistics across all layers:

```
# Collect statistics for each layer
layer_stats = []
for layer_name in network.get_layers():
    result = (
        Q.nodes()
         .from_layers(L[layer_name])
         .compute("degree", "betweenness_centrality", "clustering")
         .execute(network)
    )
    df = result.to_pandas()

    layer_stats.append({
        'layer': layer_name,
        'node_count': result.count,
        'avg_degree': df['degree'].mean(),
        'max_betweenness': df['betweenness_centrality'].max(),
        'avg_clustering': df['clustering'].mean(),
    })

# Convert to DataFrame for analysis
import pandas as pd
summary_df = pd.DataFrame(layer_stats)
print(summary_df)
```

Layer Algebra for Complex Filters

Combine layers using set operations:

```
from py3plex.dsl import L

# Nodes in social OR work layers
social_or_work = (
```

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```
    Q.nodes()
     .from_layers(L["social"] + L["work"])
     .compute("degree")
     .execute(network)
)

# Nodes in social BUT NOT bots
legitimate_social = (
    Q.nodes()
     .from_layers(L["social"] - L["bots"])
     .compute("degree")
     .execute(network)
)

# Intersection of layers
overlap = (
    Q.nodes()
     .from_layers(L["layer1"] & L["layer2"])
     .execute(network)
)
```

Aggregations by Node Attributes

Group and aggregate by node properties:

```
# Compute average centrality by community
result = (
    Q.nodes()
     .compute("betweenness_centrality", "community")
     .execute(network)
)

df = result.to_pandas()
community_centrality = df.groupby('community')['betweenness_centrality'].agg(['mean',
↪'std', 'max'])
print(community_centrality)
```

Result Conversion

The DSL provides flexible export capabilities for different analysis workflows.

To Pandas DataFrames

Convert query results to pandas for statistical analysis:

```
# Execute query
result = (
    Q.nodes()
     .from_layers(L["social"])
     .compute("degree", "betweenness_centrality", "clustering")
     .execute(network)
)
```

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```
# Convert to DataFrame
df = result.to_pandas()

# Standard pandas operations
print(df.describe())
print(df[df['degree'] > 10])
df.to_csv("results.csv", index=False)
```

DataFrame structure:

- Each row is a node
- Columns include: node_id, layer, computed measures

To NetworkX

Export filtered nodes as a NetworkX subgraph:

```
# Query for high-degree nodes
result = (
    Q.nodes()
     .where(degree__gt=10)
     .execute(network)
)

# Extract as NetworkX subgraph
node_ids = list(result.node_ids)
subgraph = network.core_network.subgraph(node_ids)

# Use NetworkX algorithms
import networkx as nx
communities = nx.community.louvain_communities(subgraph)
```

To CSV/JSON

Direct export without intermediate DataFrame:

```
# Export to CSV
(
    Q.nodes()
     .from_layers(L["social"])
     .compute("degree", "betweenness_centrality")
     .export_csv("social_centrality.csv")
     .execute(network)
)

# Export to JSON
(
    Q.nodes()
     .compute("degree")
     .export_json("node_degrees.json", orient="records")
```

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```
    .execute(network)
)
```

Supported formats:

- CSV (with custom delimiter)
- JSON (various orientations)
- pandas DataFrame (in-memory)

Workflow Integration

Pipeline Composition

Chain multiple queries in analysis pipelines:

```
# Step 1: Find high-degree nodes
hubs = (
    Q.nodes()
     .where(degree__gt=20)
     .execute(network)
)

# Step 2: Compute centrality only for hubs
hub_centrality = (
    Q.nodes()
     .where(node_id__in=list(hubs.node_ids))
     .compute("betweenness_centrality", "closeness_centrality")
     .execute(network)
)

# Step 3: Identify super-hubs
super_hubs = (
    Q.nodes()
     .where(
         node_id__in=list(hubs.node_ids),
         betweenness_centrality__gt=0.1
     )
     .execute(network)
)

print(f"Hubs: {hubs.count}, Super-hubs: {super_hubs.count}")
```

Combining with sklearn

Integrate DSL results with machine learning pipelines:

```
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler

# Extract features using DSL
result = (
    Q.nodes()
```

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```
    .compute("degree", "betweenness_centrality", "clustering")
    .execute(network)
)

df = result.to_pandas()

# Prepare features
features = df[['degree', 'betweenness_centrality', 'clustering']].values
scaler = StandardScaler()
features_scaled = scaler.fit_transform(features)

# Cluster nodes
kmeans = KMeans(n_clusters=5, random_state=42)
df['cluster'] = kmeans.fit_predict(features_scaled)

# Analyze clusters
print(df.groupby('cluster')[['degree', 'betweenness_centrality']].mean())
```

Custom Measures

While the DSL provides built-in measures, you can compute custom measures using pandas:

```
# Get base measures
result = (
    Q.nodes()
     .compute("degree", "betweenness_centrality")
     .execute(network)
)

df = result.to_pandas()

# Compute custom measure
df['influence_score'] = (
    0.6 * df['degree'] / df['degree'].max() +
    0.4 * df['betweenness_centrality'] / df['betweenness_centrality'].max()
)

# Find top nodes by custom measure
top_influential = df.nlargest(10, 'influence_score')
print(top_influential[['node_id', 'influence_score']])
```

Performance Tips

Query Optimization

For large-scale dynamics simulations:

- Use fewer replicates during development, scale up for production
- Consider using numpy backend for faster computation
- Track only essential measures to reduce memory usage
- Use smaller time steps when necessary but be aware of computational cost

Memory Management

For very large networks or long simulations:

- Avoid tracking full trajectories (`trajectory` measure) if not needed
- Use measure-specific exports rather than full result objects
- Process results in batches for very long time series

Summary

This chapter covered:

1. **DSL-based dynamics simulation** — Declarative framework for SIS, SIR, and Random Walk
2. **Mathematical formalism** — Formal definitions of epidemic models
3. **Multilayer coupling** — Simulating dynamics across network layers
4. **Query integration** — Using Q.nodes() for targeted initial conditions
5. **Parameter comparison** — Systematic exploration of parameter space
6. **Result analysis** — Rich result objects with pandas export and plotting

Key takeaways:

- The dynamics DSL follows the same design philosophy as the query DSL
- Simulations are fully declarative and composable
- Integration with query DSL enables sophisticated initial condition specification
- Results are analysis-ready with pandas, xarray, and plotting support

Further Reading

- Introduction to the Py3plex DSL
- The Builder API and Explain Plans
- `examples/network_analysis/example_dsl_dynamics.py` — Complete examples
- `examples/advanced/example_dynamics_core.py` — Core dynamics classes
- `docfiles/sir_epidemic_simulator.rst` — SIR multiplex simulator documentation

1.7.11 Limitations and Stability Guarantees

This chapter explicitly marks feature status for user confidence

DSL Feature Status

Stable APIs

Features with strong backward-compatibility guarantees:

- **Node queries** — `Q.nodes()` with filtering and measures
- **Layer algebra** — Union, difference, intersection operations
- **Core measures** — degree, betweenness, closeness, pagerank
- **Result export** — pandas, JSON, CSV formats

Guarantee: Stable APIs will not break in minor version updates (1.x → 1.y).

Experimental Features

Functional but may change in future versions:

- **Edge queries** — `Q.edges()` (limited implementation)
- **Advanced aggregations** — `GROUP BY` equivalents
- **Nested subqueries** — Complex query composition

Note: Experimental features are marked in documentation. Use with awareness that API may evolve.

Planned Features

Roadmap items (briefly mentioned, not detailed):

- **Full edge query support** — Complete parity with node queries
- **Graph pattern matching** — Motif detection DSL
- **Temporal queries** — Time-aware filtering

[Keep this section SHORT—no long speculative lists]

Current Limitations

Query Capabilities

What the DSL **can** do:

- Node filtering by degree, layer, and computed measures
- Compute centrality measures
- Layer algebra operations
- Export results in multiple formats

What the DSL **cannot** (yet) do:

- Complex edge queries (partial support only)
- Nested subqueries
- Aggregate functions (SUM, AVG, etc.)
- Join operations across layers

Performance Boundaries

- **Small networks** (<10k nodes) — No issues
- **Medium networks** (10k-100k nodes) — Most queries fast
- **Large networks** (>100k nodes) — Some queries may be slow; use filtering early

Scale and Performance

Query Complexity

[Table showing O(n), O(n log n), O(n^2) operations]

Memory Requirements

[Guidelines for large networks]

When Not to Use the DSL

Use direct NetworkX/NumPy for:

- Single-layer graphs (DSL overhead not needed)
- Custom algorithms requiring fine-grained control
- Performance-critical inner loops

API Versioning Policy

Semantic Versioning

py3plex follows semantic versioning (MAJOR.MINOR.PATCH):

- **MAJOR** (1.x → 2.x) — Breaking changes possible
- **MINOR** (1.1 → 1.2) — New features, stable API unchanged
- **PATCH** (1.1.0 → 1.1.1) — Bug fixes only

Deprecation Policy

Deprecated features are:

1. Marked in documentation and code (warnings)
2. Maintained for at least one minor version
3. Removed in next major version

Migration Guides

[Commit to providing migration guides for breaking changes]

Summary

Stable and production-ready:

- Node queries with filtering
- Core centrality measures
- Layer algebra
- Result export

Use with awareness:

- Experimental features may change
- Large network performance depends on query structure
- Some features are planned but not yet implemented

[Focus on transparency and user confidence]

Source files: - docfiles/user_guide/dsl.rst (limitations sections) - docfiles/algorithm_roadmap.rst (for planned features—keep brief)

1.7.12 Case Study 1 — Social Multiplex Network

Work in Progress

This case study provides a template and outline for a complete social multiplex network analysis. The workflow structure is production-ready, but uses placeholder data. Adapt this template to your own datasets.

DSL in Case Studies

This case study demonstrates DSL throughout the analysis workflow:

```
from py3plex.dsl import Q, L

# 1. Quick exploratory analysis
for platform in ["facebook", "twitter", "linkedin"]:
    stats = (
        Q.nodes()
         .from_layers(L[platform])
         .compute("degree", "clustering")
         .execute(network)
    )
    df = stats.to_pandas()
    print(f"{platform}: avg degree = {df['degree'].mean():.2f}")

# 2. Find cross-platform influencers
influencers = (
    Q.nodes()
     .from_layers(L["facebook"] + L["twitter"] + L["linkedin"])
     .where(degree__gt=50)
     .compute("betweenness_centrality")
     .order_by("-betweenness_centrality")
     .limit(20)
     .export_csv("influencers.csv")
     .execute(network)
)

# 3. Platform-specific analysis
twitter_only = (
    Q.nodes()
     .from_layers(L["twitter"] - L["facebook"] - L["linkedin"])
     .compute("degree")
     .execute(network)
)
```

DSL enables rapid iteration in exploratory research!

Domain Context

Social media users often maintain multiple online identities across platforms (Facebook, Twitter, LinkedIn, Instagram). This creates a **social multiplex network** where:

- **Nodes** represent users

- **Layers** represent platforms
- **Intra-layer edges** are friendships/follows within a platform
- **Inter-layer edges** link the same user across platforms

Research questions:

1. How do user roles vary across platforms?
2. Are influential users consistent across layers, or platform-specific?
3. Do communities align across platforms or fragment differently?
4. What cross-layer patterns emerge (e.g., Twitter influencers who are Facebook novices)?

Dataset Structure

Example dataset: Multi-platform social network

- **Nodes:** ~5,000 users (some present on multiple platforms)
- **Layers:** 3 platforms (Facebook, Twitter, LinkedIn)
- **Edges:** ~20,000 connections
- **Attributes:** User demographics (age, location), timestamps

Data format (edgelist):

```
# Format: source, source_layer, target, target_layer, weight
user_123, facebook, user_456, facebook, 1
user_123, twitter, user_789, twitter, 1
user_123, facebook, user_123, twitter, 1  # Inter-layer link
```

Loading the Data

Create and Load Network

```
from py3plex.core import multinet

# Create multilayer network
network = multinet.multi_layer_network(directed=False)

# Load from edgelist
network.load_network('social_multiplex.edgelist', input_type='edgelist')

# Verify structure
print(f"Nodes: {network.number_of_nodes()}")
print(f"Edges: {len(network.get_edges())}")
print(f"Layers: {network.get_layers()}")

# Basic statistics
network.basic_stats()
```

Expected output:

```
Nodes: 5234
Edges: 18976
Layers: ['facebook', 'twitter', 'linkedin']
```

Full Analysis Pipeline

Step 1: Exploratory Analysis

Start with layer-level statistics:

```
from py3plex.dsl import Q, L
import pandas as pd

# Collect statistics for each layer
layer_summary = []

for layer in ["facebook", "twitter", "linkedin"]:
    result = (
        Q.nodes()
         .from_layers(L[layer])
         .compute("degree", "clustering")
         .execute(network)
    )
    df = result.to_pandas()

    layer_summary.append({
        'layer': layer,
        'nodes': result.count,
        'avg_degree': df['degree'].mean(),
        'max_degree': df['degree'].max(),
        'avg_clustering': df['clustering'].mean()
    })

summary_df = pd.DataFrame(layer_summary)
print(summary_df)
```

Interpretation: Compare layer densities and clustering to understand platform differences.

Step 2: Community Detection

Find communities using multilayer Louvain:

```
from py3plex.algorithms.community_detection import multilayer_louvain

# Detect communities
communities = multilayer_louvain.best_partition(network.core_network)

# Add communities as node attributes
for node, comm_id in communities.items():
    network.core_network.nodes[node]['community'] = comm_id

# Count communities
n_communities = len(set(communities.values()))
print(f"Found {n_communities} communities")

# Largest communities
from collections import Counter
community_sizes = Counter(communities.values())
```

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```
print("Top 5 communities by size:")
for comm_id, size in community_sizes.most_common(5):
    print(f"  Community {comm_id}: {size} nodes")
```

Step 3: Centrality Analysis

Identify influential users using multilayer PageRank:

```
from py3plex.algorithms.centrality_toolkit import multilayer_pagerank

# Compute multilayer PageRank
pagerank_scores = multilayer_pagerank(network.core_network)

# Use DSL to find top influencers
result = (
    Q.nodes()
     .compute("degree", "betweenness_centrality")
     .execute(network)
)

df = result.to_pandas()
df['pagerank'] = df['node_id'].map(pagerank_scores)

# Top influencers
top_influencers = df.nlargest(20, 'pagerank')
print(top_influencers[['node_id', 'degree', 'betweenness_centrality', 'pagerank']])
```

Step 4: Cross-Layer Patterns

Analyze how user importance varies across platforms:

```
# For each user, compute degree in each layer
user_profiles = {}

for layer in ["facebook", "twitter", "linkedin"]:
    result = (
        Q.nodes()
         .from_layers(L[layer])
         .compute("degree")
         .execute(network)
    )
    df = result.to_pandas()

    for _, row in df.iterrows():
        user_id = row['node_id'][0]  # Extract base user ID
        if user_id not in user_profiles:
            user_profiles[user_id] = {}
        user_profiles[user_id][f"{layer}_degree"] = row['degree']

# Convert to DataFrame
profile_df = pd.DataFrame.from_dict(user_profiles, orient='index').fillna(0)
```

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```
# Find cross-layer imbalances
profile_df['max_degree'] = profile_df.max(axis=1)
profile_df['min_degree'] = profile_df.min(axis=1)
profile_df['imbalance'] = profile_df['max_degree'] / (profile_df['min_degree'] + 1)

# Users with high imbalance (platform specialists)
specialists = profile_df.nlargest(10, 'imbalance')
print("Platform specialists (high cross-layer imbalance):")
print(specialists)
```

Step 5: Visualization

Create publication-ready visualizations:

```
from py3plex.visualization.multilayer import draw_multilayer_default
import matplotlib.pyplot as plt

# Visualize network with communities colored
draw_multilayer_default(
    network.get_layers(),
    node_size=8,
    labels=True,
    background_shape="circle",
    scale_by_size=True,  # Scale nodes by degree
    display=True
)

plt.title("Social Multiplex Network - Community Structure")
plt.savefig("social_multiplex_communities.png", dpi=300, bbox_inches='tight')
```

Key Findings (Template)

Community Structure

Observation: [Describe community patterns found]

- **Cross-platform communities:** X% of users in same community across all layers
- **Platform-specific communities:** Y% of communities confined to single layer
- **Community alignment:** Normalized Mutual Information = Z

Interpretation: [What do these patterns mean for user behavior?]

Cross-Layer Roles

Observation: [Describe role variation across platforms]

- **Consistent influencers:** N users with high centrality on all platforms
- **Platform specialists:** M users highly central on one platform only
- **Role switching:** Average centrality correlation across layers =

Interpretation: [How do user strategies differ by platform?]

Pitfalls Encountered

Data Cleaning Issues

Challenge: User identity resolution across platforms

- Usernames differ across platforms
- Manual matching required for subset of users
- Missing links lead to underestimated cross-layer effects

Solution: Use email-based or profile-based matching when available

Performance Considerations

Challenge: Betweenness centrality computation slow for large networks

- $O(n^3)$ complexity becomes prohibitive for >10,000 nodes
- Consider sampling or approximation algorithms

Solution: Use degree or PageRank for large-scale analysis; reserve betweenness for focused subgraphs

Reproducibility

Code Repository

Full analysis code available at: `examples/case_studies/social_multiplex_analysis.py`

To reproduce this analysis:

```
# Clone repository
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex

# Install dependencies
pip install -e .

# Run analysis
python examples/case_studies/social_multiplex_analysis.py
```

Data Availability

Synthetic data for this case study is available in: `examples/datasets/synthetic_social_multiplex.edgelist`

For real datasets, consider:

- **Twitter + Facebook:** (pending data availability)
- **Academic multiplex:** DBLP + ArXiv coauthorship networks
- **Contact datasets:** See `multilayer_datasets/` directory

Summary

This case study demonstrated:

1. **Data loading** from edgelist format
2. **Exploratory analysis** using DSL for layer statistics

3. **Community detection** with multilayer Louvain
4. **Centrality analysis** with multilayer PageRank
5. **Cross-layer pattern detection** (role switching, specialists)
6. **Visualization** for publication

Key workflow:

1. Load → 2. Explore → 3. Detect communities → 4. Compute centrality → 5. Analyze patterns → 6. Visualize

Adapt this workflow to your own social multiplex datasets by:

- Adjusting layer names
- Modifying centrality thresholds
- Adding domain-specific measures
- Customizing visualizations

1.7.13 Case Study 2 — Biological Multilayer Network

Work in Progress

This case study is currently a template. The workflow demonstrates how to approach biological network analysis with py3plex, but uses placeholder data. Future editions will include a complete worked example with real biological datasets.

Domain Context

Biological systems exhibit multilayer structure through different interaction types:

- **Protein-protein interactions** — Physical binding
- **Regulatory relationships** — Gene expression control
- **Metabolic pathways** — Enzyme-substrate relationships
- **Co-expression networks** — Correlated expression across conditions

A **biological multiplex network** represents these different interaction mechanisms as separate layers while capturing the same entities (genes/proteins) across layers.

Research questions:

1. How do protein functions differ across interaction types?
2. Which genes are central in regulation but peripheral in metabolism?
3. Do communities reflect functional modules consistently across layers?
4. How do dynamics (e.g., epidemic-like spreading) differ by interaction type?

Dataset Structure (Template)

Example dataset: Multi-omic interaction network

- **Nodes:** ~2,000 genes/proteins
- **Layers:** Physical interaction, regulatory, metabolic
- **Edges:** ~10,000 interactions across layers

- **Attributes:** GO terms, tissue specificity, expression levels
- **Focus:** Dynamics simulation and layer-specific analysis

Data format:

```
# Format: gene_A, interaction_type, gene_B, interaction_type, confidence
BRCA1, protein_interaction, TP53, protein_interaction, 0.95
BRCA1, regulation, ATM, regulation, 0.87
TP53, metabolic, MDM2, metabolic, 0.92
```

Loading and Preprocessing

Create and Load Network

```
from py3plex.core import multinet

# Create biological multilayer network
network = multinet.multi_layer_network(directed=True)  # Regulation is directed

# Load from biological database format
network.load_network('biological_multiplex.edgelist', input_type='edgelist')

# Add gene annotations (GO terms, etc.)
# annotations = load_annotations('gene_annotations.tsv')
# for node in network.get_nodes():
#     network.core_network.nodes[node]['GO_terms'] = annotations.get(node[0], [])

print(f"Loaded {network.number_of_nodes()} genes")
print(f"Layers: {network.get_layers()}")
```

Analysis Pipeline Sketch

Step 1: Network Topology

Analyze structural properties of each interaction layer:

```
from py3plex.dsl import Q, L

# Compare topology across interaction types
for layer in ["protein_interaction", "regulation", "metabolic"]:
    result = (
        Q.nodes()
         .from_layers(L[layer])
         .compute("degree", "clustering")
         .execute(network)
    )
    df = result.to_pandas()
    print(f"{layer}: avg degree = {df['degree'].mean():.2f}, "
          f"avg clustering = {df['clustering'].mean():.3f}")
```

Expected patterns:

- Protein interaction: High clustering (modules)
- Regulation: Lower clustering (hierarchical)

- Metabolic: Medium clustering (pathways)

Step 2: Dynamics Simulation

Model information spreading or cascade dynamics using SIR model:

```
from py3plex.dynamics import SIRDynamics

# Simulate disease/perturbation spreading
sir = SIRDynamics(
    network,
    beta=0.3,      # Transmission rate
    gamma=0.1,     # Recovery rate
    initial_infected=0.05  # Start with 5% infected
)
sir.set_seed(42)

results = sir.run(steps=100)

# Analyze outbreak size
final_recovered = results.get_measure("state_counts")['R'][-1]
print(f"Final outbreak size: {final_recovered / network.number_of_nodes():.1%}")
```

Research question: How does perturbation spread differently through physical vs. regulatory interactions?

Step 3: Layer-Specific Analysis

Identify genes that are central in one layer but not others:

```
# Find regulatory hubs
regulatory_hubs = (
    Q.nodes()
     .from_layers(L["regulation"])
     .where(degree__gt=20)
     .compute("betweenness_centrality")
     .execute(network)
)

# Find protein interaction hubs
ppi_hubs = (
    Q.nodes()
     .from_layers(L["protein_interaction"])
     .where(degree__gt=20)
     .compute("betweenness_centrality")
     .execute(network)
)

# Compare overlap
reg_hub_ids = set(regulatory_hubs.node_ids)
ppi_hub_ids = set(ppi_hubs.node_ids)

overlap = reg_hub_ids & ppi_hub_ids
reg_only = reg_hub_ids - ppi_hub_ids
ppi_only = ppi_hub_ids - reg_hub_ids
```

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```
print(f"Overlap: {len(overlap)} genes")
print(f"Regulatory only: {len(reg_only)} genes")
print(f"PPI only: {len(ppi_only)} genes")
```

Step 4: Intervention Analysis

Simulate what-if scenarios: node removal, layer removal:

```
import copy

# Baseline dynamics
baseline_sir = SIRDynamics(network, beta=0.3, gamma=0.1)
baseline_sir.set_seed(42)
baseline_results = baseline_sir.run(steps=100)
baseline_outbreak = baseline_results.get_measure("state_counts")['R'][-1]

# Intervention: Remove top hub
network_intervened = copy.deepcopy(network)
top_hub = regulatory_hubs.node_ids[0]
network_intervened.core_network.remove_node(top_hub)

# Dynamics after intervention
intervened_sir = SIRDynamics(network_intervened, beta=0.3, gamma=0.1)
intervened_sir.set_seed(42)
intervened_results = intervened_sir.run(steps=100)
intervened_outbreak = intervened_results.get_measure("state_counts")['R'][-1]

reduction = (baseline_outbreak - intervened_outbreak) / baseline_outbreak
print(f"Outbreak reduced by {reduction:.1%} after removing hub")
```

Key Findings (Template)

Spreading Patterns

Expected observation: Cascade dynamics depend on layer structure

- **Protein interaction layer:** Slow, localized spreading (high clustering)
- **Regulatory layer:** Fast, global spreading (low clustering, directed)
- **Metabolic layer:** Intermediate spreading

Interpretation: Regulatory interactions enable rapid perturbation propagation

Layer Interactions

Expected observation: Cross-layer hub genes

- Genes highly central in both protein interaction and regulation
- These genes are critical: removing them has amplified impact
- Potential drug targets or disease genes

Summary

This case study template demonstrates:

1. **Biological network loading** with proper directionality
2. **Layer-specific topology** analysis
3. **Dynamics simulation** (SIR model for perturbation spreading)
4. **Intervention analysis** (what-if scenarios)
5. **Cross-layer hub identification**

To complete this case study:

1. Obtain biological multiplex dataset (e.g., STRING + RegulonDB + KEGG)
2. Add gene annotations (GO terms, pathways)
3. Run analysis pipeline
4. Validate findings against known biology
5. Visualize with annotations

Relevant examples:

- `examples/dynamics/example_sir_model.py` — SIR dynamics
- `examples/network_analysis/example_multilayer_centrality.py` — Centrality analysis
- `docfiles/sir_epidemic_simulator.rst` — SIR documentation

1.7.14 Case Study 3 — Transportation Network

Work in Progress

This case study is currently a placeholder. It will demonstrate advanced multilayer network analysis techniques applied to transportation systems. Future editions will include temporal analysis and large-scale DSL queries.

Domain Context

Urban transportation systems are inherently multilayer:

- **Nodes:** Locations (stations, stops, intersections)
- **Layers:** Transportation modes (subway, bus, bike-share, walking)
- **Edges:** Direct connections between locations within each mode
- **Inter-layer edges:** Transfer points between modes
- **Temporal dimension:** Schedules, rush hours, seasonal patterns

Research questions:

1. How do disruptions in one mode affect overall connectivity?
2. Which stations are critical hubs across multiple modes?
3. How do communities of well-connected areas differ by mode?
4. What is the optimal multi-modal route between locations?

Potential Analysis Directions

Resilience Analysis

Simulate mode disruptions and measure impact:

```
from py3plex.dsl import Q, L

# Baseline connectivity
baseline = (
    Q.nodes()
     .compute("betweenness_centrality")
     .execute(network)
)

# Remove subway layer
reduced_network = network.remove_layer("subway")

# Recompute connectivity
disrupted = (
    Q.nodes()
     .compute("betweenness_centrality")
     .execute(reduced_network)
)

# Identify most affected locations
# ...
```

Multi-Modal Path Analysis

Find optimal paths using multiple transportation modes:

```
# Shortest path across modes
import networkx as nx

path = nx.shortest_path(
    network.core_network,
    source=('location_A', 'bus'),
    target=('location_B', 'subway')
)

# Identify mode switches in path
# ...
```

Temporal Dynamics

Analyze how network properties change throughout the day:

```
# Load time-stamped network
# for time_window in time_windows:
#     network_t = load_network_for_time(time_window)
#     stats = compute_stats(network_t)
#     temporal_stats.append(stats)
```

DSL-Heavy Analysis

Demonstrate advanced DSL patterns:

```
from py3plex.dsl import Q, L, Param

# Parameterized cross-modal analysis
hub_query = (
    Q.nodes()
     .from_layers(
         L[Param.str("mode1")] + L[Param.str("mode2")]
     )
     .where(degree__gt=Param.int("threshold"))
     .compute("betweenness_centrality", "closeness_centrality")
     .order_by("-betweenness_centrality")
     .limit(20)
)

# Execute for different mode combinations
for mode1, mode2 in [("subway", "bus"), ("bus", "bike"), ("subway", "bike")]:
    result = hub_query.execute(
        network,
        mode1=mode1,
        mode2=mode2,
        threshold=10
    )
    print(f"{mode1} + {mode2}: {result.count} hubs")
```

Summary

This case study (when completed) will demonstrate:

1. **Temporal multilayer networks** — Time-varying structure
2. **Resilience analysis** — Impact of layer removal
3. **Multi-modal routing** — Path finding across layers
4. **Large-scale DSL queries** — Advanced query patterns
5. **Geospatial visualization** — Map-based network rendering

To complete:

1. Obtain transportation dataset (GTFS format, OpenStreetMap)
2. Construct multilayer representation
3. Implement temporal slicing
4. Run resilience experiments
5. Visualize with geospatial coordinates

Relevant resources:

- GTFS (General Transit Feed Specification) for public transit data
- OpenStreetMap for walking/cycling networks
- `examples/network_analysis/` for DSL patterns

1.7.15 Testing and Validation

py3plex uses a comprehensive testing strategy to ensure correctness across algorithms, data structures, and the DSL. This chapter provides a high-level overview of testing philosophy and organization. **For detailed validation scripts and test code, see Appendix C.**

Testing Philosophy

py3plex testing follows four principles:

1. **Correctness first:** Algorithms must produce mathematically correct results
2. **Conservation laws:** Physical constraints (e.g., probability conservation) must hold
3. **Regression prevention:** Known-good outputs are compared against current runs
4. **Property testing:** Invariants should hold for random inputs

Test Categories

- **Unit tests** — Individual functions and methods in isolation
- **Integration tests** — Module interactions and end-to-end workflows
- **Property-based tests** — Hypothesis-driven testing with random inputs
- **Regression tests** — Compare against reference runs for dynamics models

Test Organization

The test suite is organized by module and functionality:

```
tests/
├── test_core.py                  # Core data structures
├── test_dsl*.py                  # DSL functionality (10+ files)
├── test_dynamics*.py             # Dynamics models
├── test_algorithms*.py           # Algorithm correctness
├── test_centrality*.py           # Centrality measures
├── test_community*.py            # Community detection
├── test_io*.py                   # I/O and serialization
├── test_uncertainty*.py          # Uncertainty quantification
└── property/                     # Property-based tests (planned)
```

Current coverage: ~85% code coverage across core modules.

Key Validation Strategies

Random Walk Conservation

Random walks must conserve probability—transition probabilities from any state must sum to 1:

```
def test_random_walk_conservation():
    """Verify random walk conserves probability."""
    # Implementation in tests/test_random_walk_conservation.py
    # Computes transition matrix and checks row sums  1.0
```

Tests: `tests/test_paths.py` includes walk conservation checks.

Node2Vec Validation

Validate that Node2Vec biases (parameters p and q) have the intended effects:

- Higher p → reduced probability of returning to previous node
- Higher q → preference for local exploration vs. distant jumps

Tests: `tests/test_node2vec.py` validates bias behavior.

Community Detection Validation

Verify that community detection algorithms satisfy basic properties:

- **Modularity bounds:** Q [-0.5, 1.0]
- **Partition completeness:** Every node assigned to exactly one community
- **Singleton handling:** Isolated nodes form their own communities

Tests: `tests/test_community*.py` files validate Louvain, Infomap, and other algorithms.

Dynamics Validation

Epidemic models must satisfy conservation laws:

- **SIS:** S(t) + I(t) = N for all t
- **SIR:** S(t) + I(t) + R(t) = N for all t
- **Steady state:** SIS reaches equilibrium for subcritical parameters

Tests: `tests/test_dynamics.py` validates SIR, SIS, and other models with reference runs.

Running Tests

Basic Test Execution

```
# Run all tests
pytest tests/

# Run specific test file
pytest tests/test_dsl.py

# Run tests matching a pattern
pytest tests/ -k "test_community"

# Run with coverage
pytest tests/ --cov=py3plex --cov-report=html

# Verbose output
pytest tests/ -v
```

Using Makefile

```
make test           # Run all tests
make test-coverage  # Generate HTML coverage report
make test-fast      # Run only fast tests (skip slow integration tests)
```

Test markers:

- `@pytest.mark.slow` — Tests that take >5 seconds
- `@pytest.mark.integration` — End-to-end integration tests
- `@pytest.mark.hypothesis` — Property-based tests

Continuous Integration

py3plex uses GitHub Actions for automated testing on every commit and pull request.

GitHub Actions

The CI pipeline tests across:

- **Python versions:** 3.8, 3.9, 3.10, 3.11, 3.12
- **Operating systems:** Ubuntu Linux, macOS, Windows
- **Test suites:** Core, DSL, algorithms, dynamics, I/O

Build status: Tests must pass on all platforms before merging.

Coverage requirements: New code should maintain or improve coverage (target: 85%+).

CI Configuration

Detailed GitHub Actions workflows and CI configuration are covered in Appendix B.

Summary

py3plex testing ensures correctness through:

1. **Comprehensive test suite** — 200+ tests across core functionality
2. **Multiple validation strategies** — Conservation laws, property tests, reference runs
3. **Continuous integration** — Automated testing on all platforms
4. **High coverage** — 85%+ code coverage

Testing is an ongoing priority. Contributions that add tests for uncovered code or validate edge cases are especially welcome.

For detailed test scripts and validation examples, see Appendix C.

Next chapter: Reproducible environments and deployment practices

1.7.16 Reproducible Environments

Reproducibility is fundamental to scientific computing. This chapter covers practices for creating consistent, reproducible analysis environments using py3plex. **For detailed Docker configurations and deployment recipes, see Appendix B.**

Reproducibility Principles

Reproducible research requires:

1. **Isolated environments** — Avoid conflicts between projects and system packages
2. **Dependency pinning** — Lock package versions to prevent drift

3. **Seed management** — Control randomness in stochastic processes
4. **Environment documentation** — Record exact configurations used

Without these practices, analyses may produce different results on different machines or at different times, undermining scientific validity.

Environment Management

Python Virtual Environments

Use `venv` or `conda` to isolate py3plex installations:

Using venv (built-in):

```
# Create virtual environment
python3 -m venv py3plex-env

# Activate (Linux/macOS)
source py3plex-env/bin/activate

# Activate (Windows)
py3plex-env\Scripts\activate

# Install py3plex
pip install py3plex

# Deactivate when done
deactivate
```

Using conda:

```
# Create conda environment
conda create -n py3plex python=3.10

# Activate
conda activate py3plex

# Install py3plex
pip install py3plex

# Deactivate
conda deactivate
```

Recommendation: Use `venv` for simple projects, `conda` when you need non-Python dependencies (e.g., graph-tool, igraph bindings).

Dependency Pinning

Lock exact package versions to ensure reproducibility:

```
# Generate requirements file with exact versions
pip freeze > requirements.txt

# This creates entries like:
# py3plex==1.0.1
```

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```
# numpy==1.24.3
# networkx==3.1
# ...

# Reproduce environment on another machine
pip install -r requirements.txt
```

Best practices:

- Commit `requirements.txt` to version control
- Regenerate when dependencies change
- Use separate files for development (`requirements-dev.txt`) and production

Example requirements.txt structure:

```
# Core dependencies
py3plex==1.0.1
numpy>=1.22.0
scipy>=1.8.0
networkx>=2.8

# Optional features
matplotlib>=3.5.0
pandas>=1.4.0
```

Docker Containers (Overview)

Docker provides the strongest reproducibility guarantees by packaging the entire environment—OS, Python, libraries, and code—into a container.

Why Docker?

- **Complete isolation** — No system Python conflicts, no OS differences
- **Perfect reproducibility** — Identical environment across machines and time
- **Portability** — Share complete analysis environment with collaborators
- **Deployment** — Easy transition from development to production

Basic Docker Usage

```
# Build py3plex image
docker build -t py3plex:latest .

# Run analysis script
docker run --rm -v $(pwd):/workspace py3plex:latest \
    python /workspace/analysis.py

# Interactive shell
docker run --rm -it py3plex:latest bash
```

When to use Docker:

- **Multi-machine reproducibility** — Running on clusters, cloud, or different dev machines

- **Long-term archival** — Preserve exact environment for future replication
- **Production deployment** — Consistent behavior from dev to production

For detailed Dockerfile examples, docker-compose configurations, and deployment recipes, see Appendix B.

Seed Management

Controlling Randomness

Set seeds for all stochastic processes to ensure deterministic results:

```
import random
import numpy as np

# Set global seeds
random.seed(42)
np.random.seed(42)

# For dynamics models
from py3plex.dynamics import SIRDynamics

sir = SIRDynamics(network, beta=0.3, gamma=0.1)
sir.set_seed(42)  # Reproducible epidemic simulation
results = sir.run(steps=100)

# For random graph generation
import networkx as nx
G = nx.erdos_renyi_graph(100, 0.1, seed=42)
```

Seed management checklist:

1. Set seeds at the **start** of scripts before any randomness
2. Document seed values in code comments and papers
3. Use **different seeds** for different experiments (42, 43, 44, . . .)
4. Re-run with multiple seeds to assess variability

Recording Environments

Document Analysis Environments

Record environment details for publication and replication:

```
import platform
import sys
import py3plex
import numpy as np
import networkx as nx

# Print environment summary
print("Environment Information:")
print(f"  Python: {sys.version}")
print(f"  Platform: {platform.platform()}")
print(f"  py3plex: {py3plex.__version__}")
```

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```
print(f"  NumPy: {np.__version__}")
print(f"  NetworkX: {nx.__version__}")
```

Include in papers/reports:

- Python version (e.g., Python 3.10.8)
- py3plex version (e.g., 1.0.1)
- Key dependency versions (NumPy, NetworkX, SciPy)
- Operating system (Linux, macOS, Windows)
- Hardware details for performance-critical analyses

Version Control for Code

Use Git to track code changes:

```
# Initialize repository
git init
git add analysis.py requirements.txt
git commit -m "Initial analysis code"

# Tag important versions
git tag -a v1.0 -m "Version used for paper submission"
```

Best practices:

- Commit `requirements.txt` and environment configs
- Use meaningful commit messages
- Tag versions used for publications
- Don't commit large datasets (use Git LFS or external storage)

Version Control for Data

For input datasets:

- **Small datasets (<100 MB):** Commit directly to Git
- **Medium datasets (100 MB - 1 GB):** Use Git LFS
- **Large datasets (>1 GB):** Store externally (Zenodo, OSF, Dataverse) and include download script

Example download script:

```
# download_data.py
import urllib.request

# Dataset DOI and URL
DOI = "10.5281/zenodo.1234567"
URL = "https://zenodo.org/record/1234567/files/network.edgelist"

print(f"Downloading dataset from {DOI}")
urllib.request.urlretrieve(URL, "data/network.edgelist")
print("Download complete")
```

Summary

Essential practices for reproducible py3plex analyses:

1. **Use virtual environments** (venv or conda) to isolate dependencies
2. **Pin exact versions** with `pip freeze > requirements.txt`
3. **Set random seeds** for all stochastic processes
4. **Record environment details** in papers and code comments
5. **Use version control** (Git) for code, Git LFS or external storage for data
6. **Consider Docker** for maximum reproducibility and deployment

For production deployments:

- Use Docker containers for complete environment isolation
- See Appendix B for Dockerfile templates and docker-compose configurations
- See Appendix A for repository structure best practices

Next chapter: Overview of the py3plex GUI for visual exploration

1.7.17 The Py3plex GUI (Overview)

The py3plex GUI is a web-based interface for interactive multilayer network exploration. It provides point-and-click access to py3plex features without requiring programming. **For detailed deployment configurations and production setup, see Appendix B.**

Status: Experimental

The py3plex GUI is **experimental** and designed for **local use only**. It is not hardened for public internet deployment. Use for exploration and learning, not production analysis.

What is the Py3plex GUI?

The GUI is a FastAPI + SvelteKit web application that provides:

- **Network upload** — Load networks from files (edgelist, GraphML, JSON)
- **Interactive visualization** — D3.js-powered network rendering
- **Query interface** — Execute DSL queries via web forms
- **Analysis workflows** — Common operations (centrality, communities) without coding
- **Export results** — Download analysis outputs as CSV/JSON

Target users: Researchers learning py3plex, educators demonstrating concepts, non-programmers exploring networks.

Architecture

Technology Stack

- **Backend:** FastAPI (Python) — Fast, async HTTP server
- **Frontend:** SvelteKit (JavaScript) — Reactive UI framework
- **Visualization:** D3.js (static), Plotly (interactive)
- **API:** RESTful HTTP endpoints for network operations

- **Storage:** Local filesystem for uploads (no database)

Component structure:

```
gui/
├── api/              # FastAPI backend
│   ├── app.py        # Main server
│   ├── routes/       # API endpoints
│   └── models/       # Data models
├── frontend/         # SvelteKit app
│   ├── src/
│   │   ├── routes/   # Pages
│   │   └── lib/      # Components
│   └── static/       # Assets
└── uploads/          # Temporary file storage
```

For detailed architecture documentation, see: `docfiles/gui/gui_architecture.rst`

Running Locally

Installation

Install py3plex with GUI dependencies:

```bash
# Install with GUI extras
pip install 'py3plex[gui]'

# Or from source
cd py3plex
pip install -e '.[gui]'
```

Quickstart

```bash
# Navigate to GUI directory
cd gui

# Start backend server
python api/app.py

# In another terminal, start frontend (development mode)
cd frontend
npm install
npm run dev

# Open browser to http://localhost:5173
```

The GUI will be available at the displayed URL (typically `http://localhost:5173` for dev mode).

Configuration

Basic configuration in `gui/api/config.py`:

```python
# Development settings
DEBUG = True
```

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```
HOST = '0.0.0.0'
PORT = 8000

# File handling
UPLOAD_FOLDER = './uploads'
MAX_UPLOAD_SIZE = 100 * 1024 * 1024  # 100 MB

# Analysis limits
MAX_NODES = 10000  # Prevent performance issues
```

Key Workflows

Upload and Explore

Basic workflow:

1. **Upload network** — Click "Upload Network", select file (edgelist, GraphML, etc.)
2. **View statistics** — See node count, edge count, layer info
3. **Visualize** — Interactive network rendering with zoom/pan
4. **Layer selection** — Toggle layers on/off for focused viewing
5. **Export** — Download visualization as PNG or SVG

Query Interface

Execute DSL queries without writing code:

1. Navigate to "Query" tab
2. Build query using form:
   - Select target: nodes or edges
   - Choose layers
   - Add filters (degree, centrality, etc.)
   - Select measures to compute
   - Set ordering and limits
3. Click "Execute"
4. View results in table
5. Export as CSV or JSON

Example: "Find top 20 nodes by betweenness centrality in the social layer"

Analysis Pipelines

Pre-built workflows for common analyses:

- **Community Detection** — Run Louvain or Infomap, visualize communities
- **Centrality Analysis** — Compute degree, betweenness, PageRank
- **Layer Comparison** — Compare statistics across layers
- **Ego Networks** — Extract neighborhoods around selected nodes

Each pipeline provides a guided interface with sensible defaults.

Limitations and Safety

Security Considerations

WARNING: The GUI is **NOT secure for public deployment**.

- **No authentication** — Anyone with URL access can use it
- **No rate limiting** — Vulnerable to resource exhaustion
- **No input sanitization** — File uploads not sandboxed
- **No HTTPS** — Data transmitted in plaintext

Safe use cases:

- Local machine only (`localhost`)
- Trusted network (lab/office LAN)
- Single-user exploration

Unsafe use cases:

- Public internet exposure
- Multi-tenant environments
- Sensitive data processing

For production deployment: See Appendix B for authentication, reverse proxy, and security hardening.

Scalability Limits

Performance degrades with network size:

- **<1,000 nodes** — Full interactivity, all features work well
- **1,000-5,000 nodes** — Some slowness, simplify visualizations
- **5,000-10,000 nodes** — Limited features, use minimal layouts
- **>10,000 nodes — Use CLI/library instead**, GUI becomes unusable

Tips for large networks:

- Use layer filtering to reduce size
- Disable interactive visualization
- Export to CSV and analyze programmatically

When to Use the GUI vs CLI

Use GUI for:

- **Exploratory analysis** — Quick network inspection and hypothesis testing
- **Learning** — Understanding py3plex features interactively
- **Demonstration** — Presenting to non-programmers or stakeholders
- **Rapid prototyping** — Testing analysis ideas before writing code

Use CLI/library for:

- **Large networks** — >5,000 nodes
- **Production workflows** — Automated, reproducible pipelines
- **Complex analyses** — Chaining multiple operations
- **Performance-critical tasks** — Maximum speed and control
- **Reproducibility** — Version-controlled scripts

Summary

The py3plex GUI provides:

- **Interactive exploration** without requiring programming skills
- **Visual network rendering** with D3.js and Plotly
- **Query interface** for executing DSL queries via forms
- **Pre-built workflows** for common analyses
- **Local deployment** for individual researchers and small teams

Key limitations:

- **Experimental status** — Not production-ready
- **Security** — Local use only, no public deployment
- **Performance** — Limited to small/medium networks (<5,000 nodes)
- **Features** — Advanced capabilities require CLI/library

For deployment details and security hardening, see Appendix B. For GUI API reference, see: `docfiles/gui/gui_api_reference.rst`

1.7.18 Appendix A: Repository Layout and Scripts

This appendix provides a reference for the py3plex repository structure and maps book examples to actual scripts.

Repository Structure

Top-Level Organization

```
py3plex/
├── py3plex/              # Main package
├── tests/                # Test suite
├── examples/             # 50+ example scripts
├── docfiles/             # Documentation source (Sphinx)
├── book/                 # This book's source files
├── gui/                  # Web GUI application
├── benchmarks/           # Performance benchmarks
├── datasets/             # Sample datasets
├── pyproject.toml        # Package configuration
├── Makefile              # Development automation
└── README.md             # Project overview
```

Main Package Structure

```
py3plex/
├── __init__.py
├── core/                 # Data structures
│   ├── multinet.py       # multi_layer_network class
│   └── random_generators.py
├── algorithms/           # Analysis algorithms
│   ├── statistics/
│   ├── centrality/
│   ├── community_detection/
│   └── paths/
├── dynamics/             # Dynamical processes
│   ├── models.py         # SIS, SIR, SEIR
│   ├── core.py
│   └── compartmental.py
├── dsl/                  # Query language
│   ├── __init__.py
│   ├── builder.py        # Builder API
│   ├── executor.py
│   └── export.py
├── visualization/        # Plotting
├── io/                   # I/O handlers
│   ├── readers.py
│   └── writers.py
├── cli.py                # Command-line interface
├── graph_ops.py          # Dplyr-style API
├── pipeline.py           # sklearn-style pipelines
└── workflows.py          # Config-driven workflows
```

Examples Directory

The `examples/` directory contains 50+ working scripts organized by topic:

```
examples/
├── getting_started/      # Introductory examples
├── network_analysis/     # Core analysis examples
│   ├── example_dsl_builder_api.py       # Chapter 8-10
│   ├── example_dsl_queries.py
│   ├── example_community_detection.py   # Chapter 7
│   └── example_centrality.py
├── dynamics/             # Epidemic and process models
├── visualization/        # Plotting examples
├── communities/          # Community detection
├── io_and_data/          # Data loading
├── cli/                  # Command-line usage
├── workflows/            # Complete workflows
└── advanced/             # Advanced topics
```

Mapping Book Chapters to Examples

Chapter 4 (Installation and Getting Started)

- `examples/getting_started/quickstart.py`
- `examples/getting_started/first_network.py`

Chapter 5 (Data Loading)

- `examples/io_and_data/load_edgelist.py`
- `examples/io_and_data/arrow_parquet_io.py`
- `examples/io_and_data/csv_loading.py`

Chapter 6 (Visualization)

- `examples/visualization/basic_plot.py`
- `examples/visualization/hairball_plot.py`
- `examples/visualization/matrix_visualization.py`

Chapter 7 (Core Algorithms)

- `examples/network_analysis/example_community_detection.py`
- `examples/network_analysis/example_centrality.py`
- `examples/network_analysis/example_explainable_centrality.py`
- `examples/dynamics/sir_epidemic.py`

Chapters 8-10 (DSL)

- `examples/network_analysis/example_dsl_builder_api.py` — **Primary reference**
- `examples/network_analysis/example_dsl_queries.py`
- `examples/network_analysis/example_dsl_advanced.py`
- `examples/network_analysis/example_dsl_community_detection.py`

Chapter 17 (GUI)

- `gui/app.py` — Main application
- `gui/README.md` — Setup instructions

Key Scripts Reference

Makefile Targets

The `Makefile` provides common development tasks:

```
make setup           # Create virtual environment
make dev-install     # Install with dev dependencies
make test            # Run test suite
make test-coverage   # Generate coverage report
```

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```
make lint            # Run code quality checks
make format          # Format code with black
make docs            # Build documentation
make clean           # Clean build artifacts
```

Development Scripts

```
scripts/
├── run_tests.py              # Test runner
├── generate_docs.py          # Documentation generator
└── benchmark.py              # Performance benchmarking
```

Configuration Files

Package Configuration

pyproject.toml — Modern Python package configuration (PEP 518):

- Package metadata (name, version, authors)
- Dependencies and extras (`[infomap]`, `[viz]`, etc.)
- Build system configuration
- Tool configurations (black, ruff, mypy)

Testing Configuration

pytest.ini — Test framework configuration:

- Test discovery patterns
- Coverage settings
- Markers for test categories

Documentation Configuration

docfiles/conf.py — Sphinx documentation configuration:

- Extensions (autodoc, napoleon, mathjax)
- Theme settings (sphinx_rtd_theme)
- Build options

Docker Configuration

Dockerfile — Container image definition **docker-compose.yml** — Multi-service orchestration

[Detailed Docker configs in Appendix B]

Data Files

Sample Datasets

```
datasets/
├── karate_multiplex.edgelist
├── example_biological.graphml
└── README.md
```

External Datasets

The `multilayer_datasets/` directory contains links and scripts for downloading larger datasets used in case studies.

Summary

Key directories:

- `py3plex/` — Main package code
- `tests/` — Test suite
- `examples/` — 50+ working examples mapped to book chapters
- `docfiles/` — Documentation source
- `gui/` — Web interface

Development workflow:

1. Clone repository
2. `make setup` to create environment
3. `make dev-install` to install dependencies
4. Modify code, run `make test`
5. Use `examples/` as reference

Finding code:

- Algorithms → `py3plex/algorithms/`
- DSL → `py3plex/dsl/`
- Examples → `examples/` (organized by topic)

1.7.19 Appendix B: Docker, Docker Compose, and Deployment

This appendix provides complete Docker configurations and deployment instructions for py3plex, including the GUI.

Dockerfile Reference

Main Dockerfile

The repository includes a Dockerfile for containerized execution:

```
# File: Dockerfile
FROM python:3.10-slim

WORKDIR /app

# Install system dependencies
RUN apt-get update && apt-get install -y \
    git \
```

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```
    build-essential \
    && rm -rf /var/lib/apt/lists/*

# Copy requirements
COPY pyproject.toml .
COPY README.md .

# Install py3plex
RUN pip install --no-cache-dir -e .

# Copy application code
COPY py3plex/ ./py3plex/
COPY examples/ ./examples/

# Set up entry point
ENTRYPOINT ["python", "-m", "py3plex"]
CMD ["--help"]
```

Building the Image

```
# Build image
docker build -t py3plex:latest .

# Build with specific Python version
docker build --build-arg PYTHON_VERSION=3.11 -t py3plex:3.11 .

# Build with tag
docker build -t py3plex:1.0 .
```

Running Containers

```
# Run command
docker run --rm py3plex:latest --version

# Run analysis script
docker run --rm py3plex:latest python examples/getting_started/quickstart.py

# Mount data directory
docker run --rm -v $(pwd)/data:/data py3plex:latest python analysis.py

# Interactive shell
docker run --rm -it py3plex:latest /bin/bash
```

Docker Compose

GUI Deployment

```
# File: docker-compose.yml
version: '3.8'

```

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```yaml
services:
  gui:
    build:
      context: .
      dockerfile: gui/Dockerfile
    ports:
      - "5000:5000"
    volumes:
      - ./data:/app/data
      - ./uploads:/app/uploads
    environment:
      - FLASK_ENV=production
      - SECRET_KEY=${SECRET_KEY}
    restart: unless-stopped

  # Optional: nginx reverse proxy
  nginx:
    image: nginx:alpine
    ports:
      - "80:80"
      - "443:443"
    volumes:
      - ./nginx.conf:/etc/nginx/nginx.conf:ro
      - ./ssl:/etc/nginx/ssl:ro
    depends_on:
      - gui
    restart: unless-stopped
```

Running with Docker Compose

```bash
# Start services
docker compose up -d

# View logs
docker compose logs -f gui

# Stop services
docker compose down

# Rebuild after changes
docker compose up -d --build
```

Production Deployment

Security Hardening

1. Use environment variables for secrets:

```bash
# .env file (DO NOT commit to git)
SECRET_KEY=your-strong-secret-key-here
DATABASE_URL=postgresql://user:pass@host/db
```

```
# docker-compose.yml
services:
  gui:
    env_file:
      - .env
```

2. Run as non-root user:

```
# Add to Dockerfile
RUN useradd -m -u 1000 py3plex
USER py3plex
```

3. Use read-only file systems where possible:

```
services:
  gui:
    read_only: true
    tmpfs:
      - /tmp
      - /var/tmp
```

Nginx Reverse Proxy

nginx.conf:

```
events {
    worker_connections 1024;
}

http {
    upstream gui {
        server gui:5000;
    }

    server {
        listen 80;
        server_name yourdomain.com;

        # Redirect to HTTPS
        return 301 https://$server_name$request_uri;
    }

    server {
        listen 443 ssl;
        server_name yourdomain.com;

        ssl_certificate /etc/nginx/ssl/cert.pem;
        ssl_certificate_key /etc/nginx/ssl/key.pem;

        # Security headers
        add_header X-Frame-Options "SAMEORIGIN";
        add_header X-Content-Type-Options "nosniff";
        add_header X-XSS-Protection "1; mode=block";
```

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```
        location / {
            proxy_pass http://gui;
            proxy_set_header Host $host;
            proxy_set_header X-Real-IP $remote_addr;
            proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
            proxy_set_header X-Forwarded-Proto $scheme;
        }

        # File upload limits
        client_max_body_size 100M;
    }
}
```

TLS/SSL Configuration

Using Let's Encrypt:

```
# Install certbot
docker run --rm -v $(pwd)/ssl:/etc/letsencrypt \
    certbot/certbot certonly --standalone \
    -d yourdomain.com \
    --email your@email.com \
    --agree-tos
```

Self-signed certificates (development only):

```
# Generate self-signed certificate
openssl req -x509 -newkey rsa:4096 \
    -keyout ssl/key.pem \
    -out ssl/cert.pem \
    -days 365 -nodes
```

Authentication Setup

Basic HTTP authentication (simple):

```
server {
    ...

    location / {
        auth_basic "Restricted Access";
        auth_basic_user_file /etc/nginx/.htpasswd;

        proxy_pass http://gui;
    }
}
```

```
# Generate .htpasswd file
htpasswd -c .htpasswd username
```

OAuth2 Proxy (advanced):

Use `oauth2-proxy` as an additional service for integration with GitHub, Google, etc.

Security Checklist

Before Public Deployment

- [ ] Use HTTPS (TLS/SSL) for all connections
- [ ] Set strong `SECRET_KEY` environment variable
- [ ] Enable authentication (HTTP Basic, OAuth2, etc.)
- [ ] Run containers as non-root user
- [ ] Use read-only file systems where possible
- [ ] Set appropriate file upload limits
- [ ] Enable security headers in nginx
- [ ] Regularly update base images (`docker pull python:3.10-slim`)
- [ ] Monitor logs for suspicious activity
- [ ] Implement rate limiting (via nginx or application)
- [ ] Sanitize user inputs (especially file uploads)
- [ ] Restrict file types for uploads
- [ ] Scan uploaded files for malware
- [ ] Use network isolation (Docker networks)
- [ ] Keep dependencies up to date (`pip install --upgrade`)

Monitoring

```
# Add health checks to docker-compose.yml
services:
  gui:
    healthcheck:
      test: ["CMD", "curl", "-f", "http://localhost:5000/health"]
      interval: 30s
      timeout: 10s
      retries: 3
```

Cloud Deployment

AWS Deployment

Using ECS (Elastic Container Service):

1. Push image to ECR (Elastic Container Registry)
2. Create ECS task definition
3. Deploy to ECS cluster
4. Use ALB (Application Load Balancer) for routing

Using EC2:

1. Launch EC2 instance

2. Install Docker
3. Copy docker-compose.yml to instance
4. Run `docker compose up -d`

Azure Deployment

Using Azure Container Instances:

```
az container create \
    --resource-group myResourceGroup \
    --name py3plex-gui \
    --image py3plex:latest \
    --cpu 2 --memory 4 \
    --ports 5000
```

GCP Deployment

Using Cloud Run:

1. Push image to Google Container Registry
2. Deploy to Cloud Run
3. Configure domain and SSL

Performance Tuning

Container Resource Limits

```
services:
  gui:
    deploy:
      resources:
        limits:
          cpus: '2.0'
          memory: 4G
        reservations:
          cpus: '1.0'
          memory: 2G
```

Caching

Use Docker layer caching to speed up builds:

```
# Install dependencies first (cached layer)
COPY pyproject.toml .
RUN pip install -e .

# Copy code second (changes frequently)
COPY py3plex/ ./py3plex/
```

Multi-Stage Builds

For smaller production images:

```
# Build stage
FROM python:3.10 as builder
WORKDIR /app
COPY pyproject.toml .
RUN pip install --user -e .

# Runtime stage
FROM python:3.10-slim
WORKDIR /app
COPY --from=builder /root/.local /root/.local
COPY py3plex/ ./py3plex/
ENV PATH=/root/.local/bin:$PATH
```

Troubleshooting

Common Issues

Issue: Container exits immediately

Check logs:

```
docker logs <container_id>
```

Issue: Permission denied on mounted volumes

Fix permissions:

```
chmod -R 777 ./data
```

Or run as correct user ID.

Issue: Out of memory

Increase Docker memory limits or container resources.

Summary

This appendix provided:

- Complete Dockerfile and docker-compose.yml examples
- Security hardening guidelines
- Production deployment configurations (nginx, TLS, authentication)
- Cloud deployment options (AWS, Azure, GCP)
- Performance tuning strategies

Key recommendations for production:

1. Always use HTTPS
2. Enable authentication
3. Run as non-root
4. Monitor and log

5. Keep dependencies updated
6. Follow security checklist

[For high-level GUI overview → Chapter 17]

1.7.20 Appendix C: Detailed Validation Scripts

This appendix provides detailed validation strategies and test scripts for ensuring correctness of multilayer network algorithms. These complement the high-level overview in Chapter 15.

Validation Philosophy

py3plex uses multiple validation strategies:

1. **Unit tests** — Test individual functions in isolation
2. **Property-based tests** — Test invariants that should always hold
3. **Reference runs** — Compare against known-good results
4. **Conservation tests** — Verify physical/mathematical laws (e.g., probability conservation)

Random Walk Validation

Conservation Tests

Random walks must conserve probability—the sum of transition probabilities from any state must equal 1:

```
# File: tests/test_random_walk_conservation.py
import pytest
import numpy as np
from py3plex.algorithms.paths import random_walk

def test_probability_conservation():
    """Verify random walk conserves probability."""
    network = create_test_network()

    # Run random walk
    walks = random_walk(network, num_walks=1000, walk_length=100)

    # Compute transition matrix
    trans_matrix = compute_transition_matrix(walks)

    # Check row sums = 1
    row_sums = trans_matrix.sum(axis=1)
    assert np.allclose(row_sums, 1.0, atol=1e-6)
```

Bias Validation

Node2Vec introduces biases (return parameter p, in-out parameter q). Validate that biases have the intended effect:

```
# File: tests/test_node2vec_bias.py
def test_return_bias():
    """Higher p should reduce probability of returning to previous node."""
    network = create_test_network()
```

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```
    # Low p (encourage return)
    walks_low_p = node2vec_walk(network, p=0.25, q=1.0, num_walks=1000)
    return_rate_low = count_immediate_returns(walks_low_p)

    # High p (discourage return)
    walks_high_p = node2vec_walk(network, p=4.0, q=1.0, num_walks=1000)
    return_rate_high = count_immediate_returns(walks_high_p)

    assert return_rate_high < return_rate_low
```

Reference Runs

Store reference outputs for deterministic tests:

```
# File: tests/test_dynamics_reference_runs.py
REFERENCE_DATA = {
    'sir_small_graph_42': {
        'beta': 0.3,
        'gamma': 0.1,
        'steps': 100,
        'seed': 42,
        'expected_final_recovered': 0.75
    }
}

def test_sir_matches_reference():
    """Ensure SIR dynamics match reference run."""
    ref = REFERENCE_DATA['sir_small_graph_42']

    network = make_tiny_chain_graph()
    sir = SIRDynamics(network, beta=ref['beta'], gamma=ref['gamma'])
    sir.set_seed(ref['seed'])

    results = sir.run(steps=ref['steps'])
    final_recovered = results.get_measure('state_counts')[2][-1] / network.number_of_
↪nodes()

    assert abs(final_recovered - ref['expected_final_recovered']) < 0.01
```

Community Detection Validation

Modularity Bounds

Modularity Q must be in range [-0.5, 1.0]:

```
# File: tests/test_community_bounds.py
def test_modularity_bounds():
    """Modularity must be in valid range."""
    network = create_test_network()

    communities = multilayer_louvain.best_partition(network.core_network)
    Q = compute_modularity(network, communities)
```

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```
    assert -0.5 <= Q <= 1.0
```

Partition Quality

Every node must be assigned to exactly one community:

```
def test_partition_completeness():
    """Every node must be in exactly one community."""
    network = create_test_network()
    communities = multilayer_louvain.best_partition(network.core_network)

    # All nodes accounted for
    assert set(communities.keys()) == set(network.get_nodes())

    # No overlapping assignments (in non-overlapping methods)
    assert len(communities) == len(set(communities.values()))
```

Singleton Detection

Validate that isolated nodes form their own communities:

```
def test_singleton_communities():
    """Isolated nodes should form singleton communities."""
    network = create_network_with_isolates()
    communities = multilayer_louvain.best_partition(network.core_network)

    isolates = [n for n in network.get_nodes() if network.core_network.degree(n) == 0]

    # Each isolate in its own community
    for node in isolates:
        community = communities[node]
        same_community = [n for n, c in communities.items() if c == community]
        assert same_community == [node]
```

Centrality Validation

Degree Centrality

Degree centrality should match actual degree counts:

```
# File: tests/test_centrality_correctness.py
def test_degree_centrality():
    """Degree centrality should match computed degrees."""
    network = create_test_network()

    import networkx as nx
    degree_cent = nx.degree_centrality(network.core_network)

    for node in network.get_nodes():
        computed = degree_cent[node]
```

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```
        expected = network.core_network.degree(node) / (network.number_of_nodes() - 1)
        assert abs(computed - expected) < 1e-10
```

Betweenness Centrality

Validate using known structures (e.g., star graphs):

```
def test_betweenness_star_graph():
    """In a star graph, center has betweenness 1.0."""
    network = create_star_graph(n=5)  # 1 center + 4 leaves

    import networkx as nx
    bc = nx.betweenness_centrality(network.core_network, normalized=True)

    center = ('center', 'layer')
    assert bc[center] == 1.0  # All paths go through center

    leaves = [n for n in network.get_nodes() if n != center]
    for leaf in leaves:
        assert bc[leaf] == 0.0  # Leaves are not on any shortest paths
```

Dynamics Validation

SIS Model Conservation

In SIS model, S(t) + I(t) = N for all t:

```
# File: tests/test_dynamics_conservation.py
def test_sis_conservation():
    """SIS model must conserve population."""
    network = create_test_network()
    N = network.number_of_nodes()

    sis = SISDynamics(network, beta=0.3, gamma=0.1)
    sis.set_seed(42)

    results = sis.run(steps=100)

    for t in range(100):
        S_t = results.get_measure('state_counts')[0][t]
        I_t = results.get_measure('state_counts')[1][t]
        assert S_t + I_t == N
```

SIR Model Conservation

In SIR model, S(t) + I(t) + R(t) = N for all t:

```
def test_sir_conservation():
    """SIR model must conserve population."""
    network = create_test_network()
    N = network.number_of_nodes()
```

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```
    sir = SIRDynamics(network, beta=0.3, gamma=0.1)
    sir.set_seed(42)

    results = sir.run(steps=100)

    for t in range(100):
        S_t = results.get_measure('state_counts')[0][t]
        I_t = results.get_measure('state_counts')[1][t]
        R_t = results.get_measure('state_counts')[2][t]
        assert S_t + I_t + R_t == N
```

Steady State Validation

SIS model should reach steady state:

```
def test_sis_steady_state():
    """SIS should reach equilibrium."""
    network = create_test_network()

    sis = SISDynamics(network, beta=0.3, gamma=0.1)
    sis.set_seed(42)

    results = sis.run(steps=1000)
    prevalence = results.get_measure('prevalence')

    # Check last 100 steps are stable
    last_100 = prevalence[-100:]
    variance = np.var(last_100)
    assert variance < 0.01  # Low variance = steady state
```

Property-Based Testing

Using Hypothesis

Property-based tests use the `hypothesis` library to generate random inputs:

```
# File: tests/property/test_paths_properties.py
from hypothesis import given, strategies as st
from hypothesis import assume

@given(
    num_nodes=st.integers(min_value=3, max_value=20),
    num_layers=st.integers(min_value=1, max_value=5),
    seed=st.integers(min_value=0, max_value=1000)
)
def test_shortest_path_length_property(num_nodes, num_layers, seed):
    """Shortest path length should be <= graph diameter."""
    network = generate_random_network(num_nodes, num_layers, seed)
    assume(nx.is_connected(network.core_network))  # Skip disconnected graphs

    # Pick random source and target
    nodes = list(network.get_nodes())
```

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```
    source, target = random.sample(nodes, 2)

    # Compute shortest path
    path = nx.shortest_path(network.core_network, source, target)
    path_length = len(path) - 1

    # Path length should be <= diameter
    diameter = nx.diameter(network.core_network)
    assert path_length <= diameter
```

Multilayer-Specific Properties

```
@given(num_nodes=st.integers(min_value=2, max_value=10))
def test_node_activity_bounds(num_nodes):
    """Node activity must be in [0, 1]."""
    network = generate_multiplex_network(num_nodes)

    for node_id in range(num_nodes):
        activity = mls.node_activity(network, str(node_id))
        assert 0.0 <= activity <= 1.0
```

Running Validation Tests

Full Test Suite

```
# Run all tests
pytest tests/

# Run only validation tests
pytest tests/ -k validation

# Run with coverage
pytest tests/ --cov=py3plex --cov-report=html
```

Property-Based Tests

```
# Run property tests (slower)
pytest tests/property/

# Run with more examples
pytest tests/property/ --hypothesis-profile=thorough
```

Continuous Validation

Tests run automatically on:

- Every commit (GitHub Actions)
- Pull requests
- Scheduled nightly runs

Summary

This appendix detailed validation strategies:

1. **Conservation tests** — Physical/mathematical laws must hold
2. **Reference runs** — Deterministic tests against known-good outputs
3. **Boundary tests** — Values must be in valid ranges
4. **Property-based tests** — Invariants hold for random inputs
5. **Structure tests** — Algorithmic correctness on known structures

Key test files:

- `tests/test_dynamics_reference_runs.py` — Reference dynamics data
- `tests/test_random_walk_conservation.py` — Probability conservation
- `tests/property/` — Property-based tests
- `tests/test_centrality_correctness.py` — Centrality validation

[For high-level testing overview → Chapter 15]

1.7.21 Appendix D: Error Handling and Exception Hierarchy

This appendix documents py3plex's exception classes and error handling conventions.

Exception Hierarchy

py3plex defines a hierarchy of exceptions for different error categories:

```
Py3plexException (base)
├── ValidationError
│   ├── InvalidNetworkError
│   ├── InvalidLayerError
│   └── InvalidNodeError
├── IOError
│   ├── FileFormatError
│   ├── FileNotFoundError
│   └── SerializationError
├── DSLError
│   ├── QuerySyntaxError
│   ├── QueryExecutionError
│   └── UnsupportedFeatureError
├── AlgorithmError
│   ├── ConvergenceError
│   └── InsufficientDataError
└── ConfigurationError
```

Base Exception

```python
class Py3plexException(Exception):
    """Base exception for all py3plex errors."""
    pass
```

All py3plex exceptions inherit from this base class, allowing users to catch any library error:

```
from py3plex.exceptions import Py3plexException

try:
    result = some_py3plex_operation()
except Py3plexException as e:
    print(f"py3plex error: {e}")
```

Validation Errors

InvalidNetworkError

Raised when a network is in an invalid state:

```
from py3plex.exceptions import InvalidNetworkError

# Example: Empty network
if network.number_of_nodes() == 0:
    raise InvalidNetworkError("Cannot compute metrics on empty network")
```

Common causes:

- Empty network (no nodes or edges)
- Disconnected graph when algorithm requires connected
- Directed graph when undirected expected (or vice versa)

InvalidLayerError

Raised when referencing a nonexistent layer:

```
from py3plex.exceptions import InvalidLayerError

if layer_name not in network.get_layers():
    raise InvalidLayerError(f"Layer '{layer_name}' does not exist")
```

Common causes:

- Typo in layer name
- Layer removed after reference created
- Case sensitivity mismatch

InvalidNodeError

Raised when referencing a nonexistent node:

```
from py3plex.exceptions import InvalidNodeError

if node not in network.core_network:
    raise InvalidNodeError(f"Node {node} not found in network")
```

I/O Errors

FileFormatError

Raised when file format is invalid or unsupported:

```
from py3plex.exceptions import FileFormatError

try:
    network.load_network("data.xyz", input_type="xyz")
except FileFormatError as e:
    print(f"Unsupported format: {e}")
```

Common causes:

- Unrecognized file extension
- Malformed file content
- Missing required fields in CSV/JSON

SerializationError

Raised during read/write operations:

```
from py3plex.exceptions import SerializationError

try:
    write(network, "output.arrow")
except SerializationError as e:
    print(f"Failed to serialize: {e}")
```

DSL Errors

QuerySyntaxError

Raised for invalid DSL query syntax:

```
from py3plex.dsl import execute_query
from py3plex.exceptions import QuerySyntaxError

try:
    result = execute_query(network, 'SELECT nodes WHRE degree > 5')  # Typo
except QuerySyntaxError as e:
    print(f"Query syntax error: {e}")
    # Suggestion: "Did you mean 'WHERE'?"
```

Common causes:

- Typos in keywords (WHRE instead of WHERE)
- Missing quotes around layer names
- Invalid comparison operators

QueryExecutionError

Raised when query execution fails:

```
from py3plex.exceptions import QueryExecutionError

try:
    result = execute_query(network, 'SELECT nodes COMPUTE nonexistent_measure')
except QueryExecutionError as e:
    print(f"Query execution failed: {e}")
```

Common causes:

- Referencing nonexistent measures
- Applying measure to unsuitable graph (e.g., eigenvector centrality on disconnected graph)
- Resource limits exceeded

UnsupportedFeatureError

Raised when using experimental or unimplemented features:

```
from py3plex.exceptions import UnsupportedFeatureError

try:
    result = Q.edges().where(weight__gt=0.5).execute(network)
except UnsupportedFeatureError as e:
    print(f"Feature not yet supported: {e}")
```

Algorithm Errors

ConvergenceError

Raised when iterative algorithms fail to converge:

```
from py3plex.exceptions import ConvergenceError

try:
    centrality = compute_eigenvector_centrality(network, max_iter=100)
except ConvergenceError as e:
    print(f"Algorithm did not converge: {e}")
```

InsufficientDataError

Raised when there's not enough data for analysis:

```
from py3plex.exceptions import InsufficientDataError

if network.number_of_nodes() < 3:
    raise InsufficientDataError("Need at least 3 nodes for community detection")
```

Configuration Errors

ConfigurationError

Raised for invalid configuration or parameters:

```
from py3plex.exceptions import ConfigurationError

if beta <= 0 or gamma <= 0:
    raise ConfigurationError("SIR parameters beta and gamma must be positive")
```

Error Handling Best Practices

Catch Specific Exceptions

Prefer catching specific exceptions over generic ones:

```
# Good: Specific exception handling
from py3plex.exceptions import InvalidLayerError

try:
    result = Q.nodes().from_layers(L["social"]).execute(network)
except InvalidLayerError:
    print("Layer 'social' not found. Available layers:", network.get_layers())

# Avoid: Generic exception (too broad)
try:
    result = Q.nodes().from_layers(L["social"]).execute(network)
except Exception as e:  # Too generic
    print(f"Something went wrong: {e}")
```

Provide Context

Include helpful context in error messages:

```
# Good: Helpful context
if layer_name not in network.get_layers():
    available = ", ".join(network.get_layers())
    raise InvalidLayerError(
        f"Layer '{layer_name}' not found. Available layers: {available}"
    )

# Less helpful
if layer_name not in network.get_layers():
    raise InvalidLayerError(f"Invalid layer")
```

Validation at Entry Points

Validate inputs early:

```
def compute_centrality(network, measure='degree'):
    """Compute node centrality."""
    # Validate early
    if network.number_of_nodes() == 0:
```

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```
        raise InvalidNetworkError("Cannot compute centrality on empty network")

    if measure not in ['degree', 'betweenness', 'closeness']:
        raise ValueError(f"Unknown measure: {measure}")

    # Proceed with computation
    ...
```

Clean Resource Cleanup

Use context managers for resources:

```
# Good: Automatic cleanup
from py3plex.io import read

try:
    graph = read('network.arrow')
    # Process graph
except IOError as e:
    print(f"Failed to read file: {e}")
finally:
    # Resources automatically cleaned up
```

Error Messages Guidelines

1. **Be specific:** State what went wrong and why
2. **Suggest fixes:** Offer alternatives when possible
3. **Include context:** Show relevant values (layer names, node counts, etc.)
4. **Be concise:** One sentence explanation, then details if needed

Examples:

```
# Good error message
raise InvalidLayerError(
    f"Layer 'socail' not found. Did you mean 'social'? "
    f"Available layers: {', '.join(network.get_layers())}"
)

# Good error message
raise InsufficientDataError(
    f"Community detection requires at least 3 nodes, but network has only "
    f"{network.number_of_nodes()}"
)

# Less helpful
raise InvalidLayerError("Layer error")  # Too vague
```

Logging

py3plex uses Python's logging module:

```
import logging

# Configure logging level
logging.basicConfig(level=logging.INFO)

# py3plex will log warnings and errors
network.load_network("data.edgelist")  # Logs any warnings
```

Available log levels:

- `DEBUG` — Detailed information for debugging
- `INFO` — General information
- `WARNING` — Recoverable issues
- `ERROR` — Serious problems
- `CRITICAL` — Fatal errors

Summary

Exception hierarchy:

- Base: `Py3plexException`
- Categories: Validation, I/O, DSL, Algorithm, Configuration

Best practices:

1. Catch specific exceptions
2. Provide helpful context in messages
3. Validate early
4. Suggest fixes when possible
5. Use logging for non-fatal issues

Example usage:

```
from py3plex.core import multinet
from py3plex.exceptions import InvalidLayerError, Py3plexException

try:
    network = multinet.multi_layer_network()
    network.load_network("data.edgelist")

    result = Q.nodes().from_layers(L["social"]).execute(network)

except InvalidLayerError as e:
    print(f"Layer not found: {e}")
except Py3plexException as e:
    print(f"py3plex error: {e}")
```

[For error recovery strategies in workflows → Chapter 10]

1.7.22 Appendix E: Extended API and DSL Reference

This appendix provides a quick reference for the py3plex API and DSL syntax.

Core API Reference

Creating Networks

```
from py3plex.core import multinet

# Create empty network
network = multinet.multi_layer_network(
    directed=False,           # True for directed networks
    network_type='multilayer' # or 'multiplex'
)
```

Adding Nodes and Edges

```
# Add edges (nodes created implicitly)
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1.0],
    ['Bob', 'friends', 'Carol', 'friends', 1.0],
], input_type="list")

# Add edges with attributes
network.add_edges([{
    'source': 'Alice',
    'source_type': 'friends',
    'target': 'Bob',
    'target_type': 'friends',
    'weight': 0.8,
    'timestamp': '2023-01-15'
}])
```

Loading Networks

```
# From file
network.load_network("data.edgelist", input_type="edgelist")

# From NetworkX
import networkx as nx
G = nx.karate_club_graph()
network.load_network_from_networkx(G, layer_name='social')
```

Basic Operations

```
# Get network information
layers = network.get_layers()
nodes = list(network.get_nodes())
edges = list(network.get_edges())

# Statistics
```

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```
network.basic_stats()
num_nodes = network.get_number_of_nodes()
num_edges = network.get_number_of_edges()

# Layer-specific
nodes_per_layer = network.get_number_of_nodes_per_layer()
layer_subgraph = network.get_layer_subgraph('friends')
```

Algorithms API

Statistics

```
from py3plex.algorithms.statistics import multilayer_statistics as mls

# Layer density
density = mls.layer_density(network, 'friends')

# Node activity (fraction of layers a node appears in)
activity = mls.node_activity(network, 'Alice')

# Layer overlap
overlap = mls.layer_overlap(network, 'friends', 'colleagues')
```

Community Detection

```
from py3plex.algorithms.community_detection import multilayer_louvain

# Detect communities
communities = multilayer_louvain.best_partition(network.core_network)

# communities = {('Alice', 'friends'): 0, ('Bob', 'friends'): 0, ...}
```

Centrality

```
import networkx as nx

# Standard centralities (use NetworkX)
G = network.core_network
degree_cent = nx.degree_centrality(G)
betweenness = nx.betweenness_centrality(G)
closeness = nx.closeness_centrality(G)

# Explainable centrality (py3plex-specific)
from py3plex.algorithms.centrality.explain import explain_node_centrality

explanation = explain_node_centrality(
    network,
    node='Alice',
    measure='degree'
)
# Returns: {'score': 4, 'layer_breakdown': {...}, ...}
```

Dynamics

```
from py3plex.dynamics.models import SIRDynamics, SISDynamics

# Configure SIR model
sir = SIRDynamics(
    network,
    beta=0.3,      # Infection rate
    gamma=0.1,     # Recovery rate
    initial_infected={'Alice': 0}  # Initial conditions
)

# Set seed for reproducibility
sir.set_seed(42)

# Run simulation
results = sir.run(steps=100)

# Extract measures
prevalence = results.get_measure('prevalence')
recovered = results.get_measure('state_counts')[2]  # State 2 = recovered
```

DSL Quick Reference

Builder API (Recommended)

```
from py3plex.dsl import Q, L, Param

# Basic query
result = Q.nodes().execute(network)

# Filter by layer
result = Q.nodes().from_layers(L["social"]).execute(network)

# Filter by condition
result = Q.nodes().where(degree__gt=5).execute(network)

# Multiple conditions
result = Q.nodes().where(layer="social", degree__gt=3).execute(network)

# Layer algebra
result = Q.nodes().from_layers(L["social"] + L["work"]).execute(network)      # Union
result = Q.nodes().from_layers(L["social"] - L["bots"]).execute(network)     # Difference
result = Q.nodes().from_layers(L["social"] & L["work"]).execute(network)     #␣
↪Intersection

# Compute measures
result = Q.nodes().compute("degree", "betweenness_centrality").execute(network)

# Order and limit
result = (
    Q.nodes()
     .compute("degree")
```

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```
    .order_by("-degree")
    .limit(10)
    .execute(network)
)

# Export results
(
    Q.nodes()
     .compute("degree")
     .export_csv("output.csv")
     .execute(network)
)

# Parameterized queries
result = (
    Q.nodes()
     .where(degree__gt=Param('min_degree'))
     .execute(network, params={'min_degree': 5})
)
```

String DSL Syntax

```
from py3plex.dsl import execute_query

# Basic syntax: SELECT target WHERE conditions COMPUTE measures

# Select all nodes
execute_query(network, 'SELECT nodes')

# Filter by layer
execute_query(network, 'SELECT nodes WHERE layer="social"')

# Filter by degree
execute_query(network, 'SELECT nodes WHERE degree > 5')

# Multiple conditions
execute_query(network, 'SELECT nodes WHERE layer="social" AND degree > 3')

# Compute measures
execute_query(network, 'SELECT nodes COMPUTE betweenness_centrality')

# Combined
execute_query(network,
    'SELECT nodes WHERE layer="social" AND degree > 3 '
    'COMPUTE degree betweenness_centrality'
)
```

DSL Operators

Comparison operators:

- = — Equal to
- != — Not equal to
- > — Greater than
- < — Less than
- >= — Greater than or equal
- <= — Less than or equal

Logical operators:

- `AND` — Logical AND
- `OR` — Logical OR
- `NOT` — Logical NOT

Filter modifiers (Builder API):

- `field=value` — Equal
- `field__ne=value` — Not equal
- `field__gt=value` — Greater than
- `field__gte=value` — Greater or equal
- `field__lt=value` — Less than
- `field__lte=value` — Less or equal

Available Measures

- `degree` — Node degree
- `degree_centrality` — Normalized degree
- `betweenness_centrality` — Betweenness
- `closeness_centrality` — Closeness
- `eigenvector_centrality` — Eigenvector
- `pagerank` — PageRank
- `clustering` — Clustering coefficient

Working with Results

```
result = Q.nodes().compute("degree").execute(network)

# Iteration
for node in result:
    print(node)

# Count
print(f"Found {result.count} nodes")
```

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```
# Measures
degrees = result.measures['degree']
for node, degree in degrees.items():
    print(f"{node}: {degree}")

# To pandas
df = result.to_pandas()
```

I/O API

Modern I/O System

```
from py3plex.io import read, write, MultiLayerGraph

# Read (auto-detects format from extension)
graph = read('network.json')
graph = read('network.arrow')
graph = read('network.parquet')

# Write
write(network, 'output.json')
write(network, 'output.arrow')
write(network, 'output.parquet')

# Specify format explicitly
graph = read('file.dat', format='json')
write(network, 'file.dat', format='arrow')
```

Supported Formats

- **JSON** — `.json` — Human-readable
- **JSONL** — `.jsonl` — Streaming
- **CSV** — `.csv` — Spreadsheet-compatible
- **Arrow** — `.arrow` — High-performance
- **Parquet** — `.parquet` — Compressed

Visualization API

Basic Visualization

```
from py3plex.visualization.multilayer import draw_multilayer_default

# Simple visualization
draw_multilayer_default([network], display=True)

# With layout options
draw_multilayer_default(
    [network],
    display=True,
```

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```
    layout='force_directed',
    node_size=300,
    edge_width=2
)
```

Matrix Visualization

```
# Visualize supra-adjacency matrix
network.visualize_matrix({"display": True})
```

Command-Line Interface

Basic Usage

```
# Show version
python -m py3plex --version

# Run self-test
python -m py3plex selftest

# Analyze network
python -m py3plex analyze network.edgelist

# Community detection
python -m py3plex communities network.edgelist --algorithm louvain
```

Summary

This appendix provided quick references for:

- Core API (creating networks, adding nodes/edges)
- Algorithms API (statistics, communities, centrality, dynamics)
- DSL syntax (both Builder API and string syntax)
- I/O API (reading/writing various formats)
- Visualization API
- Command-line interface

For detailed documentation:

- Main text chapters for concepts and examples
- Appendix A for repository layout
- Appendix D for error handling
- Online API docs: https://skblaz.github.io/py3plex/

1.7.23 Bibliography

This section provides references for further reading on multilayer networks and py3plex.

Core References

Py3plex Publications

Community Detection

Network Embeddings

Dynamics and Processes

Applications

Biological Networks

Social Networks

Transportation Networks

Software and Tools

Online Resources

Py3plex:

- Documentation: https://skblaz.github.io/py3plex/
- GitHub: https://github.com/SkBlaz/py3plex
- PyPI: https://pypi.org/project/py3plex/

Community:

- Network Science community: http://www.netscisociety.net/
- Complex Networks: https://www.complexnetworks.org/

Datasets:

- Network Repository: http://networkrepository.com/
- SNAP: Stanford Network Analysis Project: http://snap.stanford.edu/data/
- Netzschleuder: https://networks.skewed.de/

Recommended Reading Path

For Graduate Students

1. Start with [Kivelä2014] for comprehensive foundations
2. Read [Skrlj2019a] for py3plex overview
3. Read [Mucha2010] for community detection
4. Explore [Bianconi2018] textbook for deeper theory

For Practitioners

1. Start with [Skrlj2019a] for py3plex capabilities
2. Skim [Kivelä2014] for key concepts

3. Focus on application papers relevant to your domain
4. Use this book and online docs for implementation

For Researchers

1. Read [Kivelä2014] and [Boccaletti2014] thoroughly
2. Study [Domenico2013] for mathematical rigor
3. Read domain-specific application papers
4. Explore [Bianconi2018] for advanced topics

1.7.24 Citation Information

How to Cite Py3plex

If you use py3plex in your research, please cite the following paper:

BibTeX

```
@Article{Skrlj2019,
  author={Skrlj, Blaz and Kralj, Jan and Lavrac, Nada},
  title={Py3plex toolkit for visualization and analysis of multilayer networks},
  journal={Applied Network Science},
  year={2019},
  volume={4},
  number={1},
  pages={94},
  doi={10.1007/s41109-019-0203-7},
  url={https://doi.org/10.1007/s41109-019-0203-7}
}
```

For conference proceedings, you may also cite:

```
@InProceedings{Skrlj2019b,
  author="{\v{S}}krlj, Bla{\v{z}}
  and Kralj, Jan
  and Lavra{\v{c}}, Nada",
  editor="Aiello, Luca Maria
  and Cherifi, Chantal
  and Cherifi, Hocine
  and Lambiotte, Renaud
  and Li{\'o}, Pietro
  and Rocha, Luis M.",
  title="Py3plex: A Library for Scalable Multilayer Network Analysis and Visualization",
  booktitle="Complex Networks and Their Applications VII",
  year="2019",
  publisher="Springer International Publishing",
  address="Cham",
  pages="757--768",
  isbn="978-3-030-05411-3",
  doi="10.1007/978-3-030-05411-3_60"
}
```

APA Style

Škrlj, B., Kralj, J., & Lavrač, N. (2019). Py3plex toolkit for visualization and analysis of multilayer networks. *Applied Network Science*, *4*(1), 94. https://doi.org/10.1007/s41109-019-0203-7

MLA Style

Škrlj, Blaz, Jan Kralj, and Nada Lavrač. "Py3plex toolkit for visualization and analysis of multilayer networks." *Applied Network Science* 4.1 (2019): 94.

Chicago Style

Škrlj, Blaz, Jan Kralj, and Nada Lavrač. "Py3plex toolkit for visualization and analysis of multilayer networks." *Applied Network Science* 4, no. 1 (2019): 94.

Citing This Book

If you specifically reference this book (rather than the software), please cite:

```
Škrlj, B. (2025). *Practical Multilayer Network Analysis with Py3plex* (Version 1.0).
Available at: https://github.com/SkBlaz/py3plex
```

BibTeX for This Book

```
@book{Skrlj2025book,
  author = {Škrlj, Blaž},
  title = {Practical Multilayer Network Analysis with Py3plex},
  year = {2025},
  version = {1.0},
  url = {https://github.com/SkBlaz/py3plex}
}
```

Citing Specific Algorithms

If you use specific algorithms implemented in py3plex, consider citing the original papers:

Louvain Community Detection

```
@article{Blondel2008,
  title={Fast unfolding of communities in large networks},
  author={Blondel, Vincent D and Guillaume, Jean-Loup and Lambiotte, Renaud and Lefebvre,
↪ Etienne},
  journal={Journal of statistical mechanics: theory and experiment},
  volume={2008},
  number={10},
  pages={P10008},
  year={2008}
}
```

Multilayer Modularity

```
@article{Mucha2010,
  title={Community structure in time-dependent, multiscale, and multiplex networks},
  author={Mucha, Peter J and Richardson, Thomas and Macon, Kevin and Porter, Mason A and␣
↪Onnela, Jukka-Pekka},
  journal={Science},
  volume={328},
  number={5980},
  pages={876--878},
  year={2010}
}
```

Node2Vec

```
@inproceedings{Grover2016,
  title={node2vec: Scalable feature learning for networks},
  author={Grover, Aditya and Leskovec, Jure},
  booktitle={Proceedings of the 22nd ACM SIGKDD international conference on Knowledge␣
↪discovery and data mining},
  pages={855--864},
  year={2016}
}
```

Software Version Information

Reproducibility best practice is to cite the specific version of py3plex used:

```
import py3plex
print(py3plex.__version__)  # e.g., "1.0.2"
```

In your paper's methods section, include:

```
All analyses were performed using py3plex version 1.0.2 [Skrlj2019]
running on Python 3.10.
```

Or in acknowledgments:

```
This research utilized py3plex (version 1.0.2, Škrlj et al., 2019)
for multilayer network analysis.
```

License

Py3plex Core

The main py3plex library is released under the **MIT License**, a permissive open-source license that allows commercial use:

```
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Permission is hereby granted, free of charge, to any person obtaining a copy
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```

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```
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OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE
SOFTWARE.
```

Bundled Code

Important: Some bundled code (e.g., Infomap community detection in `py3plex/algorithms/community_detection/infomap/`) is licensed under **AGPLv3**, which has copyleft requirements.

If you use these features, your application may be subject to AGPLv3 requirements. See Chapter 4 for details on license considerations.

This Book

Acknowledgments

Py3plex development has been supported by:

- Slovenian Research Agency (research program P2-0103 and projects)
- Jožef Stefan Institute
- Contributors and users worldwide

The py3plex community has provided valuable feedback, bug reports, and feature suggestions. We thank all contributors.

Special thanks to:

- **NetworkX developers** — py3plex builds on NetworkX
- **Apache Arrow project** — For high-performance I/O
- **Open-source community** — For tools and libraries

Contact

For questions about citing py3plex:

- **GitHub Issues:** https://github.com/SkBlaz/py3plex/issues
- **Email:** See contact information in the repository

For research collaborations or academic inquiries, see the py3plex repository for current contact information.

CHAPTER

TWO

INDICES AND TABLES

- genindex
- modindex
- search

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Comprehensive review of multilayer network theory, mathematics, and applications. Essential reading.

[Boccaletti2014] Boccaletti, S., Bianconi, G., Criado, R., Del Genio, C. I., Gómez-Gardenes, J., Romance, M., … & Zanin, M. (2014). The structure and dynamics of multilayer networks. *Physics Reports*, 544(1), 1-122. https://doi.org/10.1016/j.physrep.2014.07.001

Physics-focused perspective on multilayer networks with emphasis on dynamics.

[Bianconi2018] Bianconi, G. (2018). *Multilayer Networks: Structure and Function*. Oxford University Press.

Comprehensive textbook covering theory, methods, and applications.

[Domenico2013] De Domenico, M., Solé-Ribalta, A., Cozzo, E., Kivelä, M., Moreno, Y., Porter, M. A., … & Arenas, A. (2013). Mathematical formulation of multilayer networks. *Physical Review X*, 3(4), 041022. https://doi.org/10.1103/PhysRevX.3.041022

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[Skrlj2019a] Škrlj, B., Kralj, J., & Lavrač, N. (2019). Py3plex toolkit for visualization and analysis of multilayer networks. *Applied Network Science*, 4(1), 94. https://doi.org/10.1007/s41109-019-0203-7

Primary py3plex paper. Cite this when using py3plex in research.

[Skrlj2019b] Škrlj, B., Kralj, J., & Lavrač, N. (2019). Py3plex: A library for scalable multilayer network analysis and visualization. In *International Conference on Complex Networks and Their Applications* (pp. 757-768). Springer. https://doi.org/10.1007/978-3-030-05411-3_60

Conference paper focusing on visualization and scalability.

[Blondel2008] Blondel, V. D., Guillaume, J. L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008(10), P10008. https://doi.org/10.1088/1742-5468/2008/10/P10008

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[Magnani2013] Magnani, M., & Rossi, L. (2013). Pareto distance for multilayer network analysis. In *Social Computing, Behavioral-Cultural Modeling and Prediction* (pp. 249-256). Springer. https://doi.org/10.1007/978-3-642-37210-0_27

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[Gallotti2014] Gallotti, R., & Barthelemy, M. (2014). Anatomy and efficiency of urban multimodal mobility. *Scientific Reports*, 4(1), 6911. https://doi.org/10.1038/srep06911

Transportation networks as multilayer systems.

[NetworkX] Hagberg, A., Swart, P., & S Chult, D. (2008). Exploring network structure, dynamics, and function using NetworkX. *Los Alamos National Lab.(LANL), Los Alamos, NM (United States)*.

NetworkX documentation: https://networkx.org/

[Arrow] Apache Arrow Development Team. (2023). Apache Arrow. https://arrow.apache.org/

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INDEX